

Lecture 11:

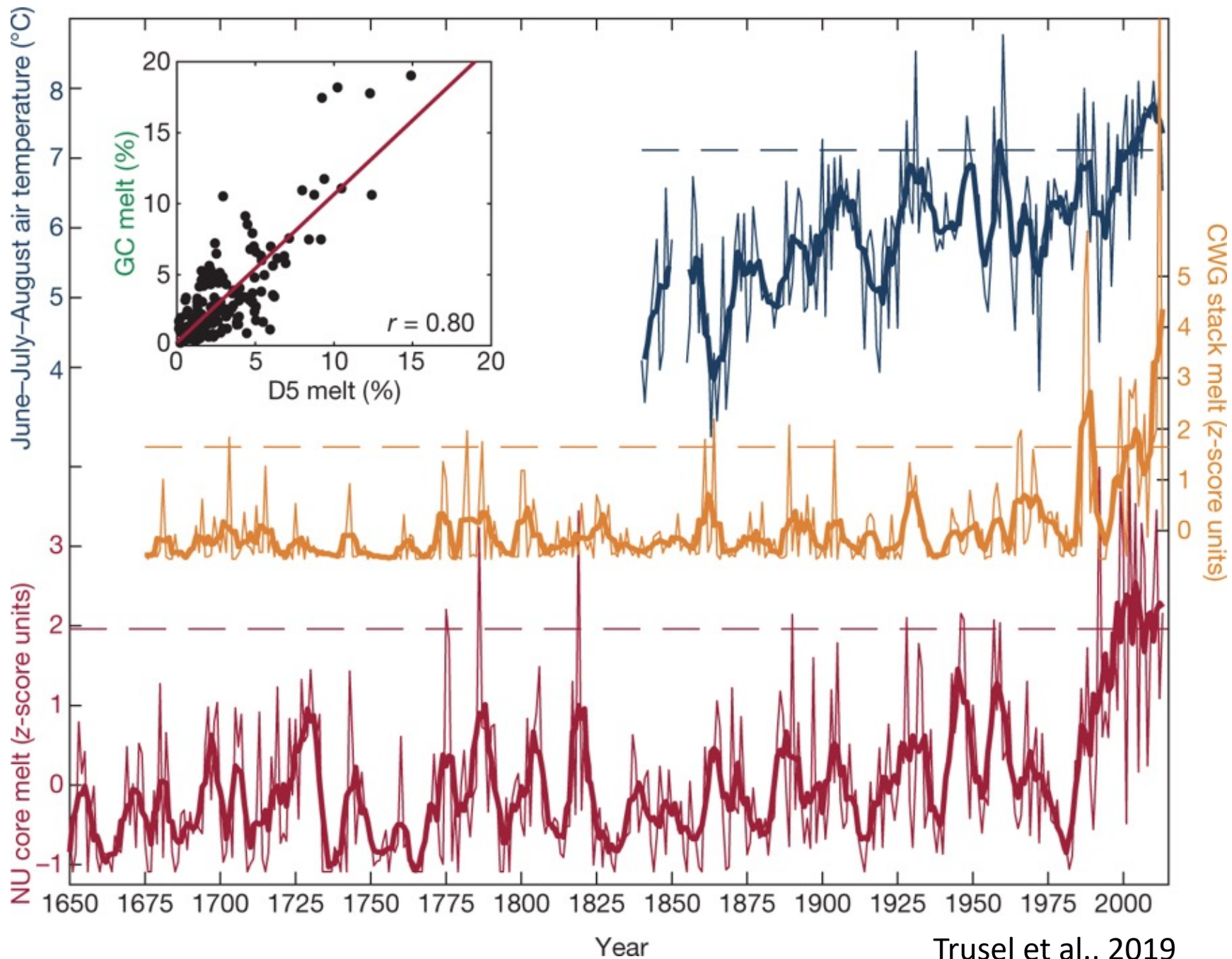
Glacial hydrology

Why?

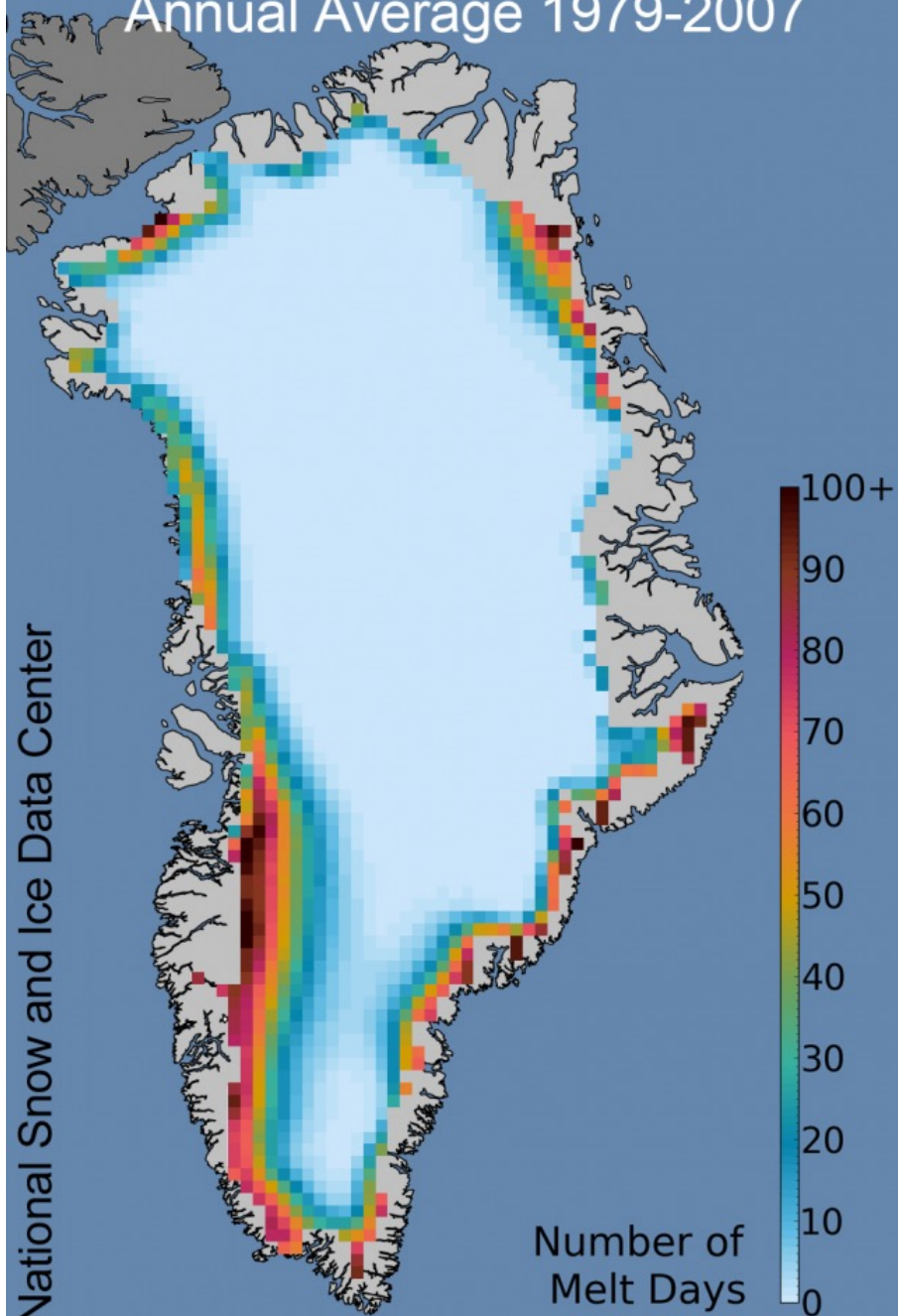
Where?

What?

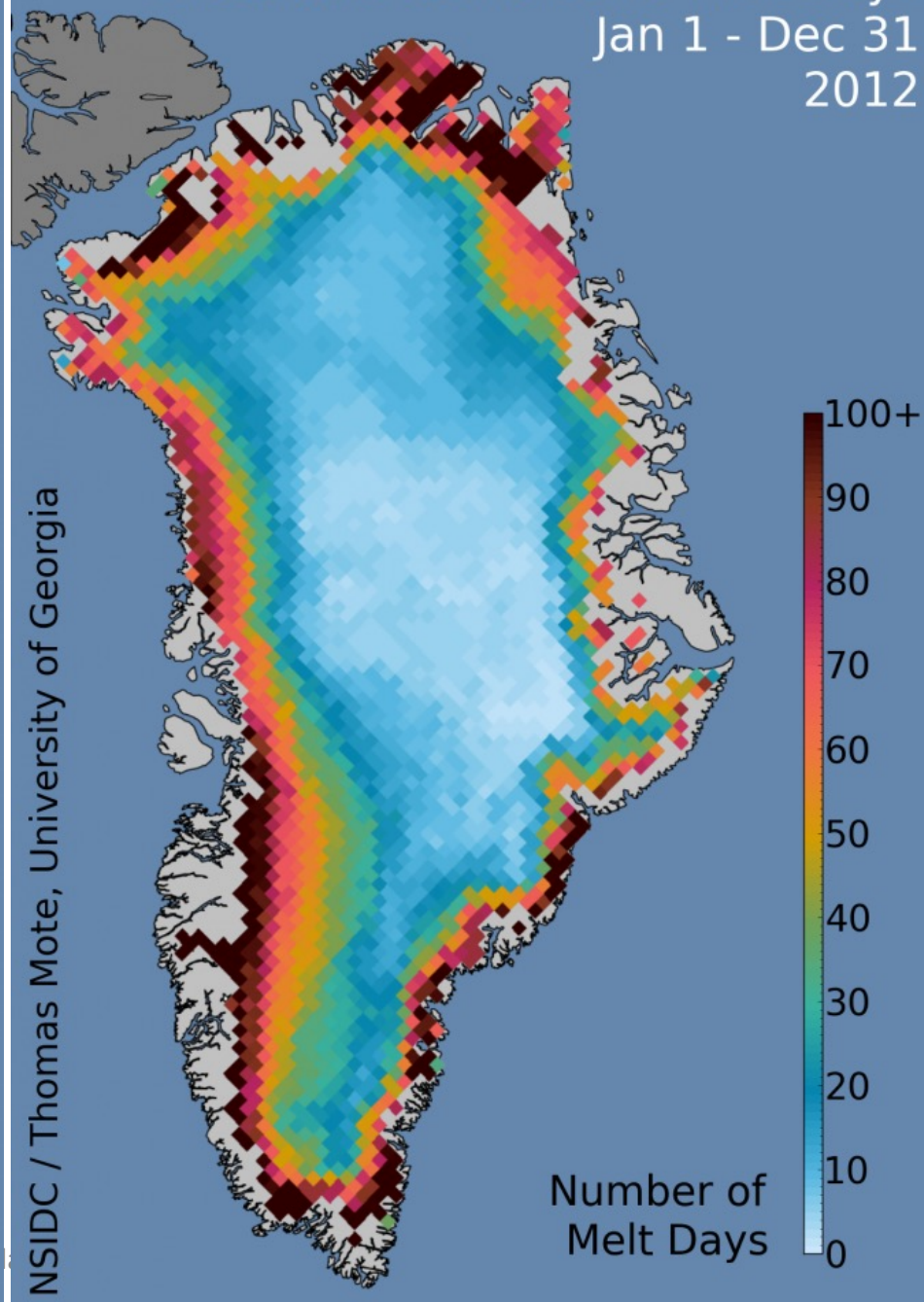
How?

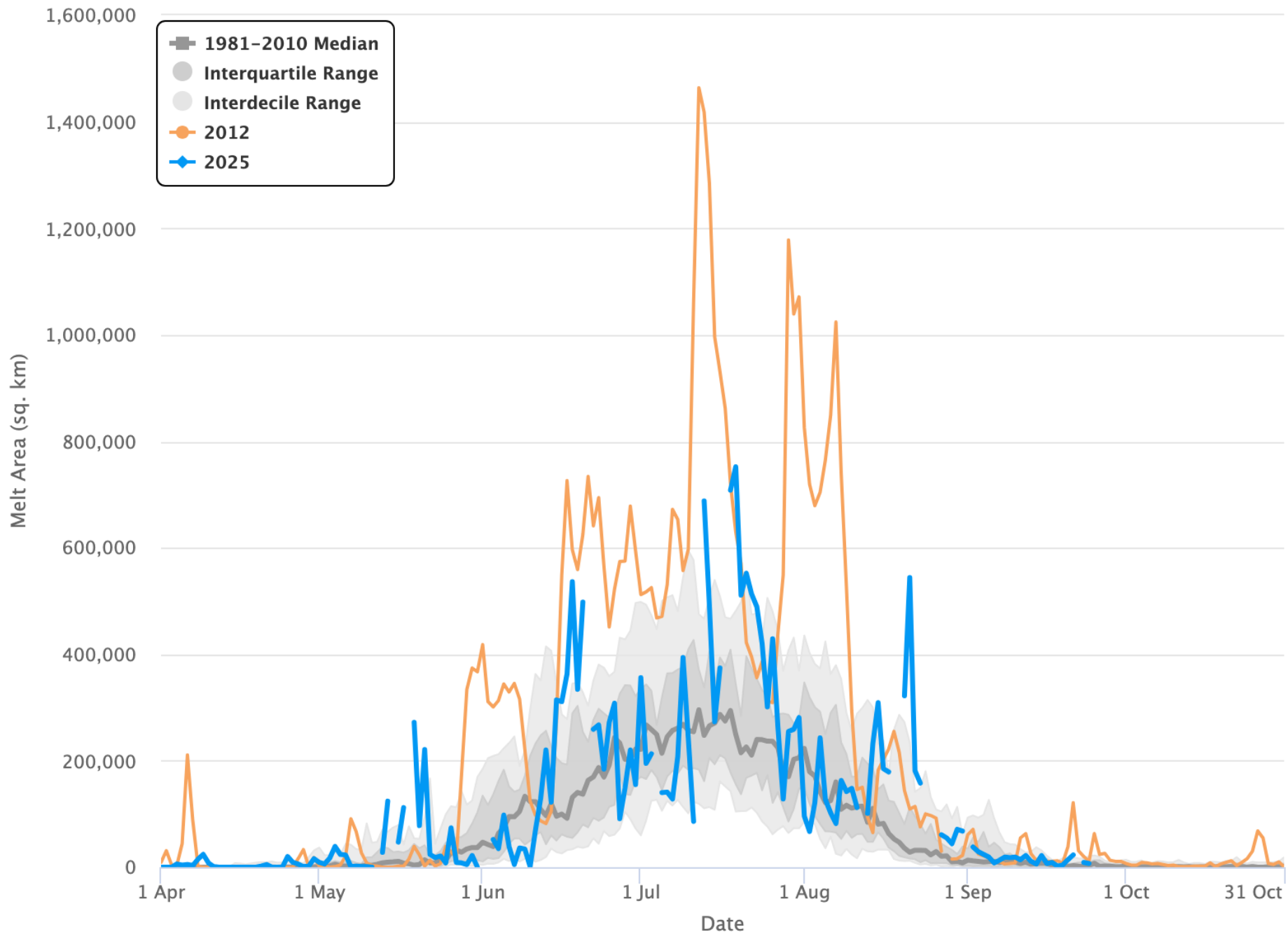


Greenland Cumulative Melt Days
Annual Average 1979-2007

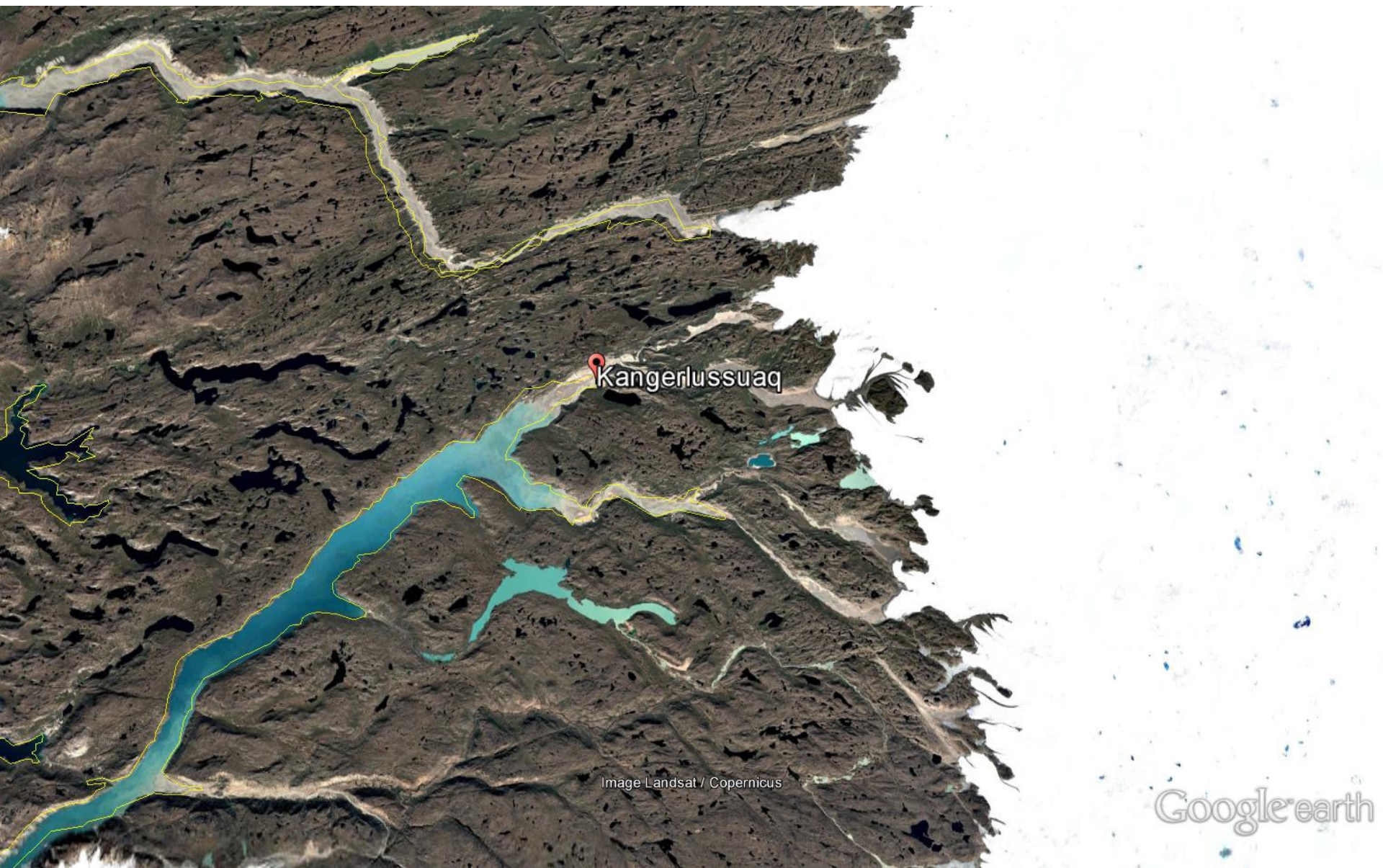


Greenland Cumulative Melt Days
Jan 1 - Dec 31
2012











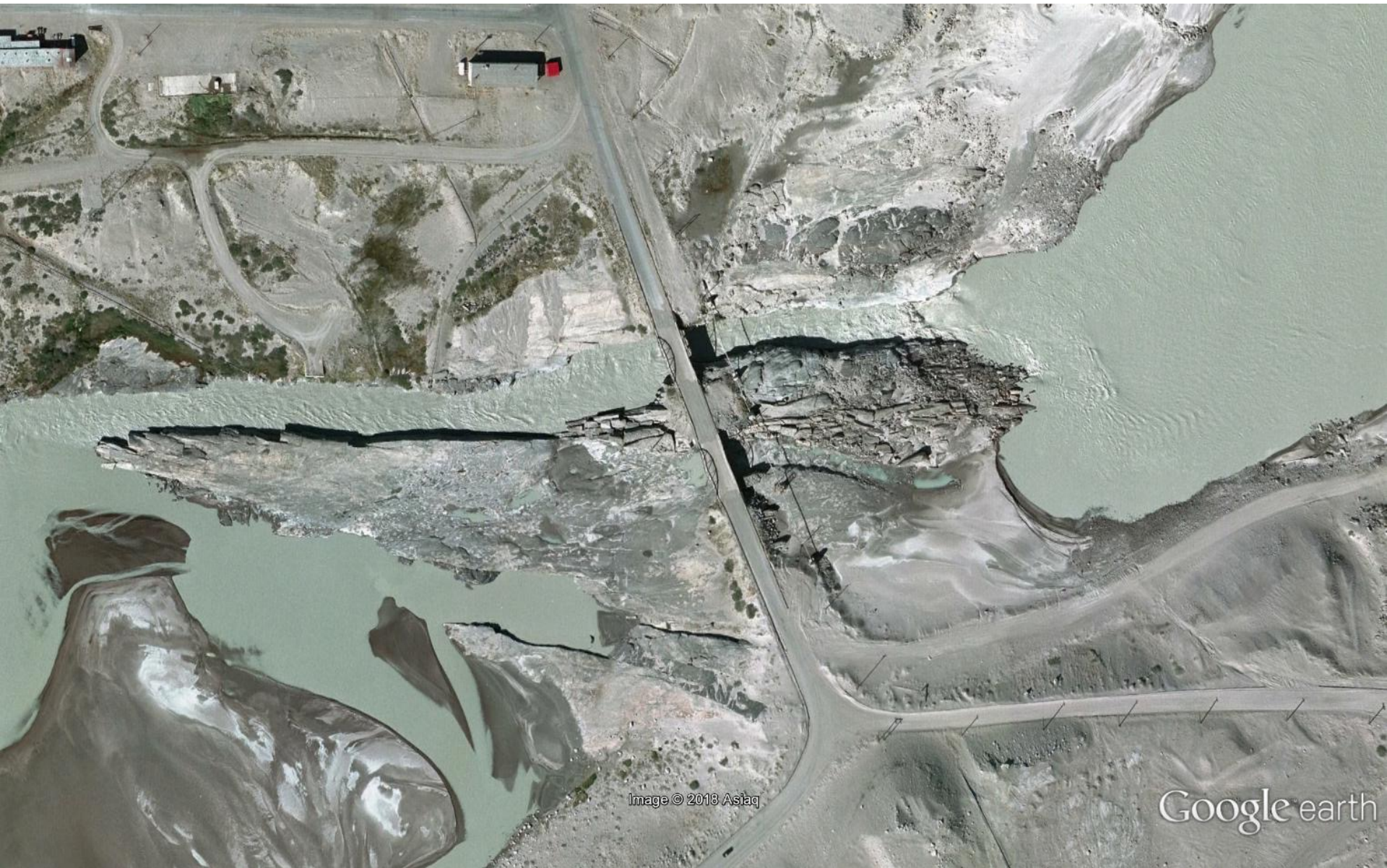


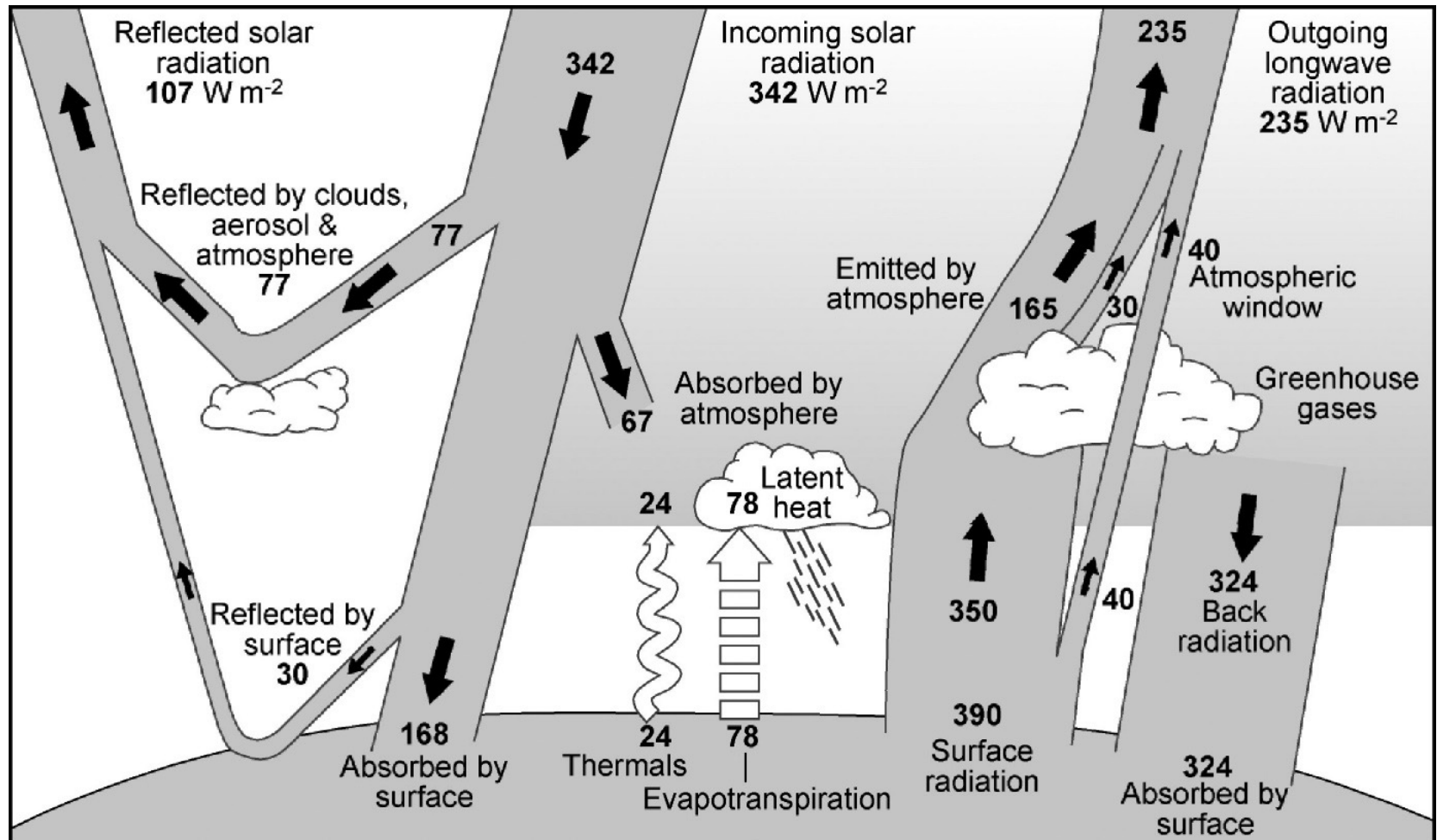
Image © 2018 Asiaq

Google earth

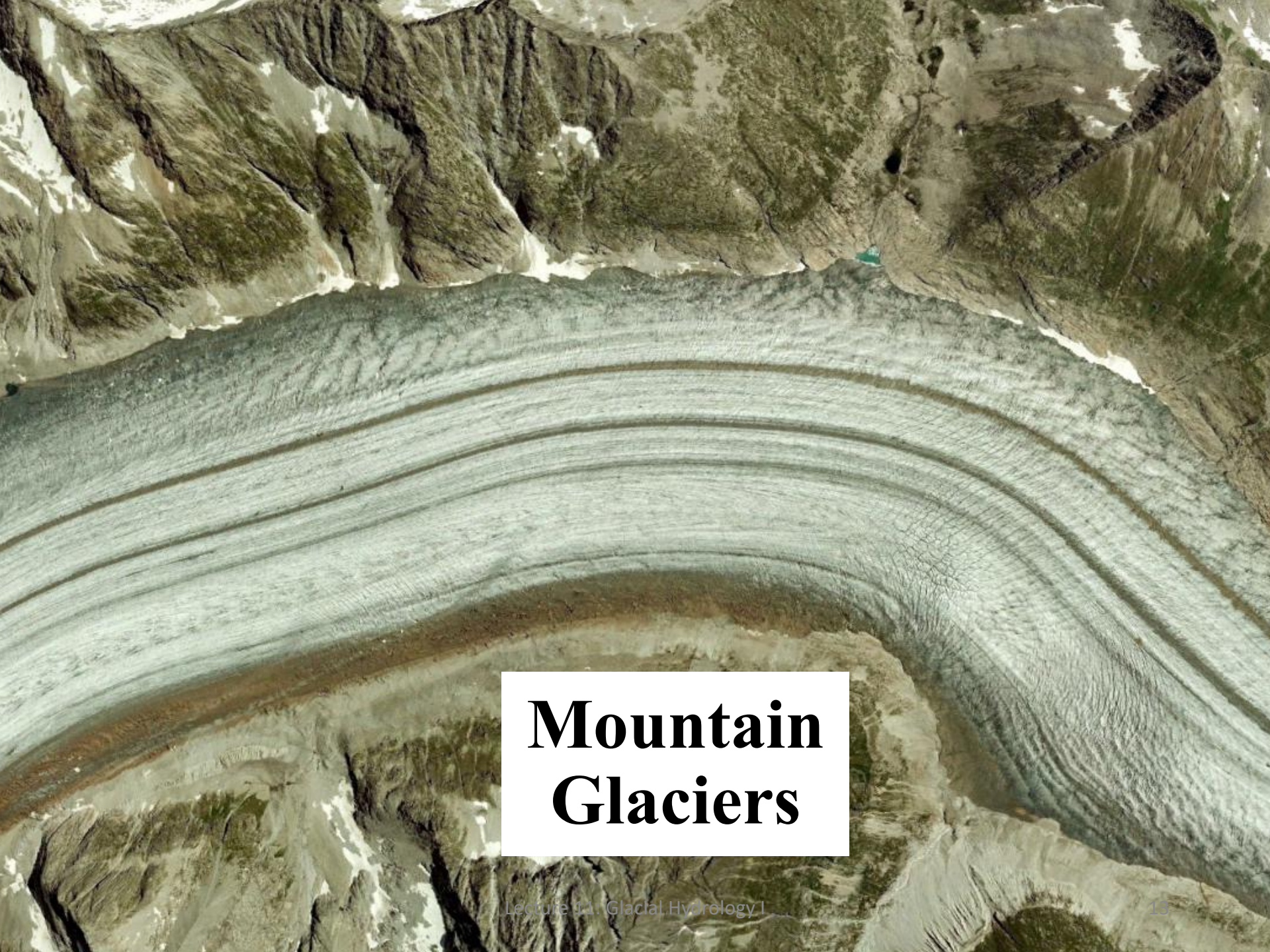


Why?

Why? – energy balance

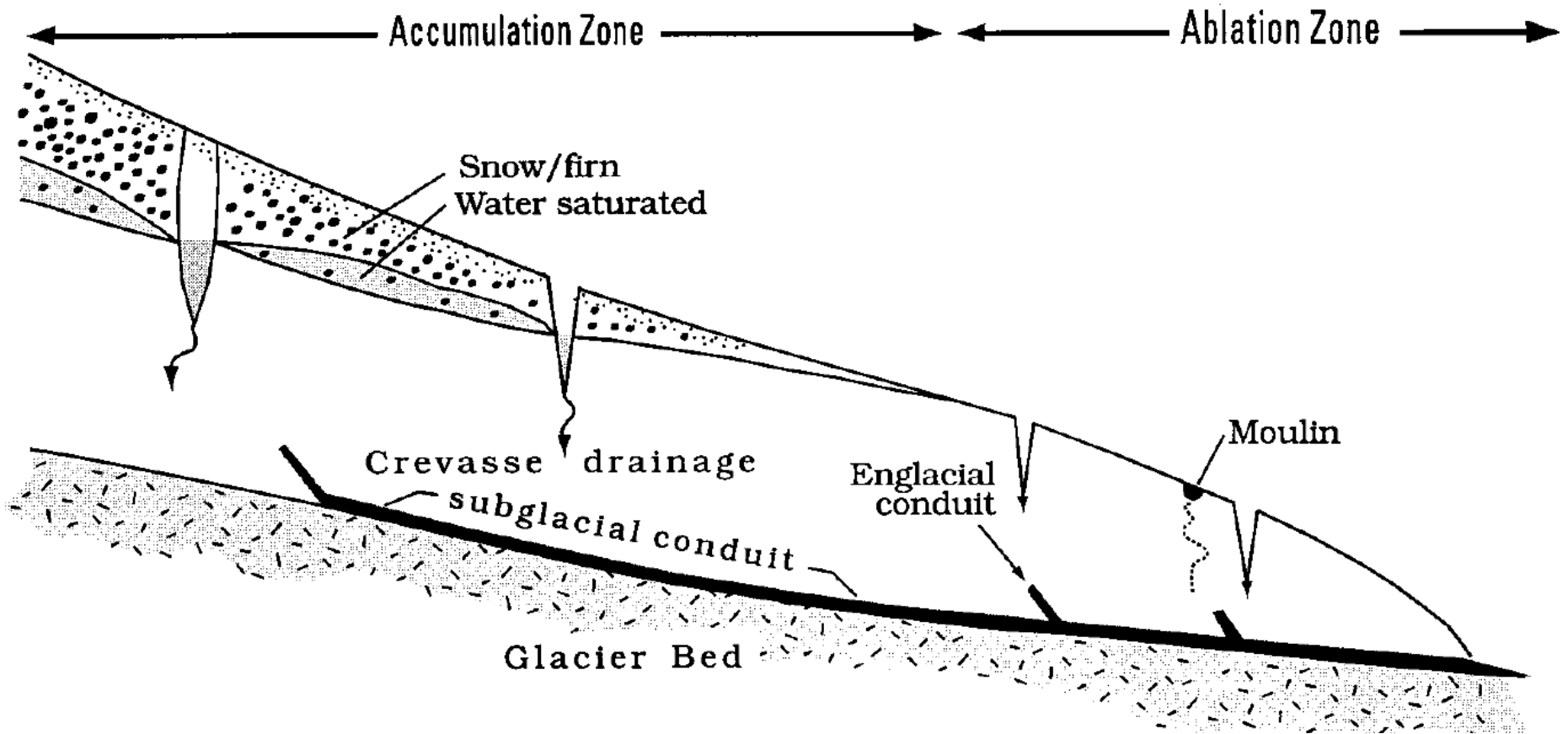


$$E_N = E_S^{\downarrow} + E_S^{\uparrow} + E_L^{\downarrow} + E_L^{\uparrow} + E_H + E_E$$



Mountain Glaciers

Where? Mountain glaciers



Fountain (1998)

Where? Mountain glaciers



Where? Mountain glaciers



Where? Mountain glaciers

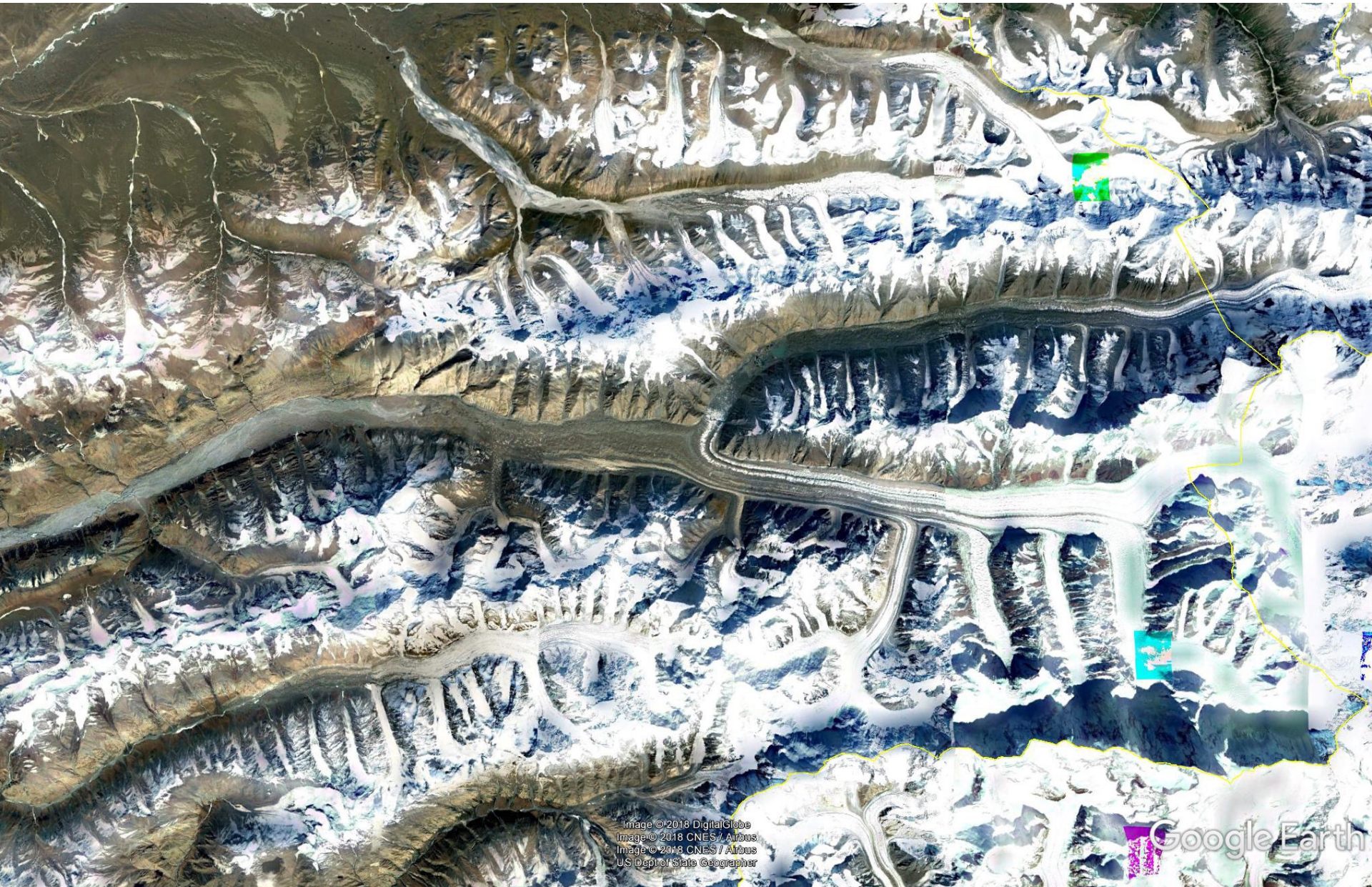


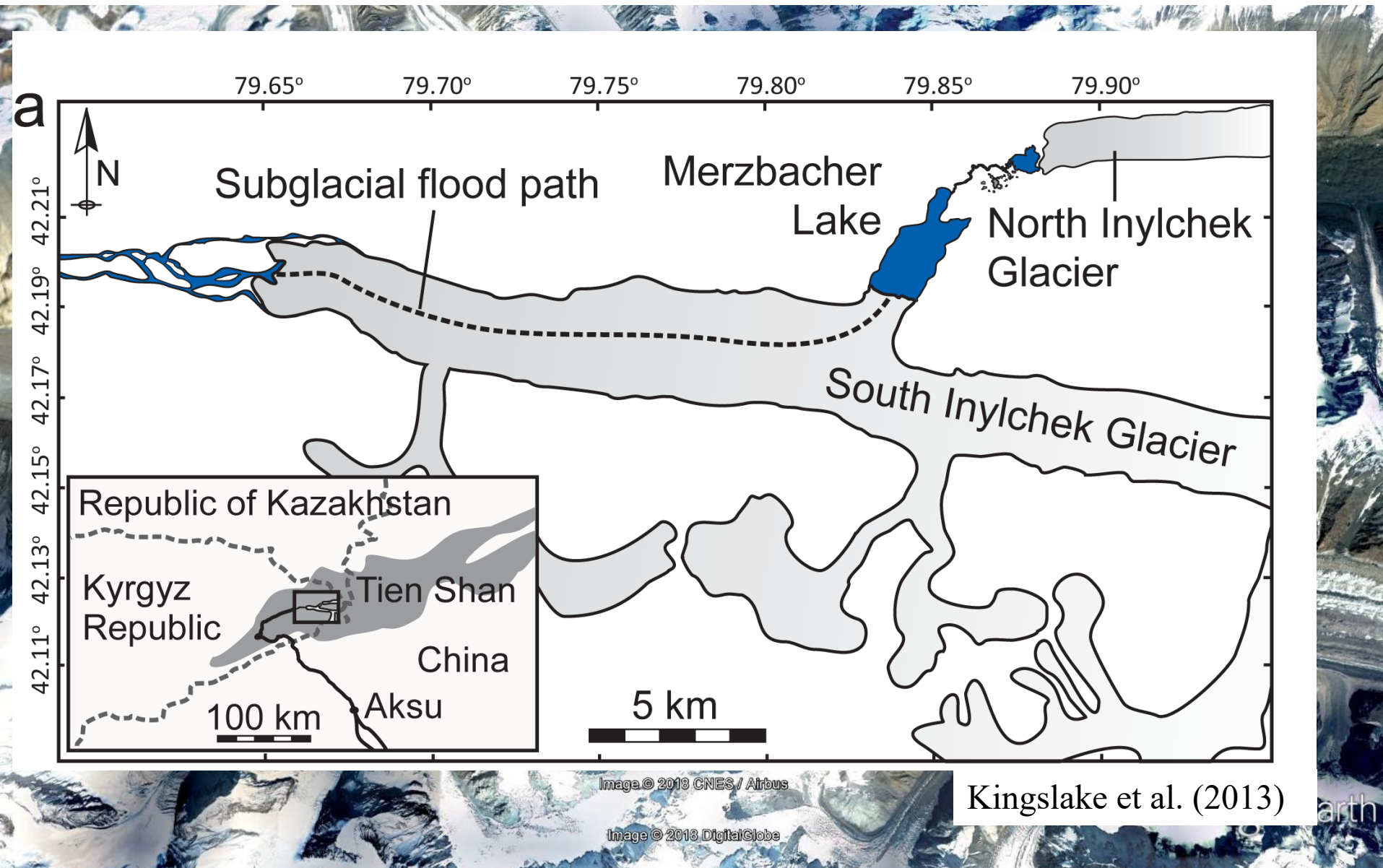
Image © 2018 DigitalGlobe
Image © 2018 CNES / Airbus
Image © 2018 CNES / Airbus
US Dept of State Geographer

Google Earth

Where? Mountain glaciers



Where? Mountain glaciers



Where? Mountain glaciers

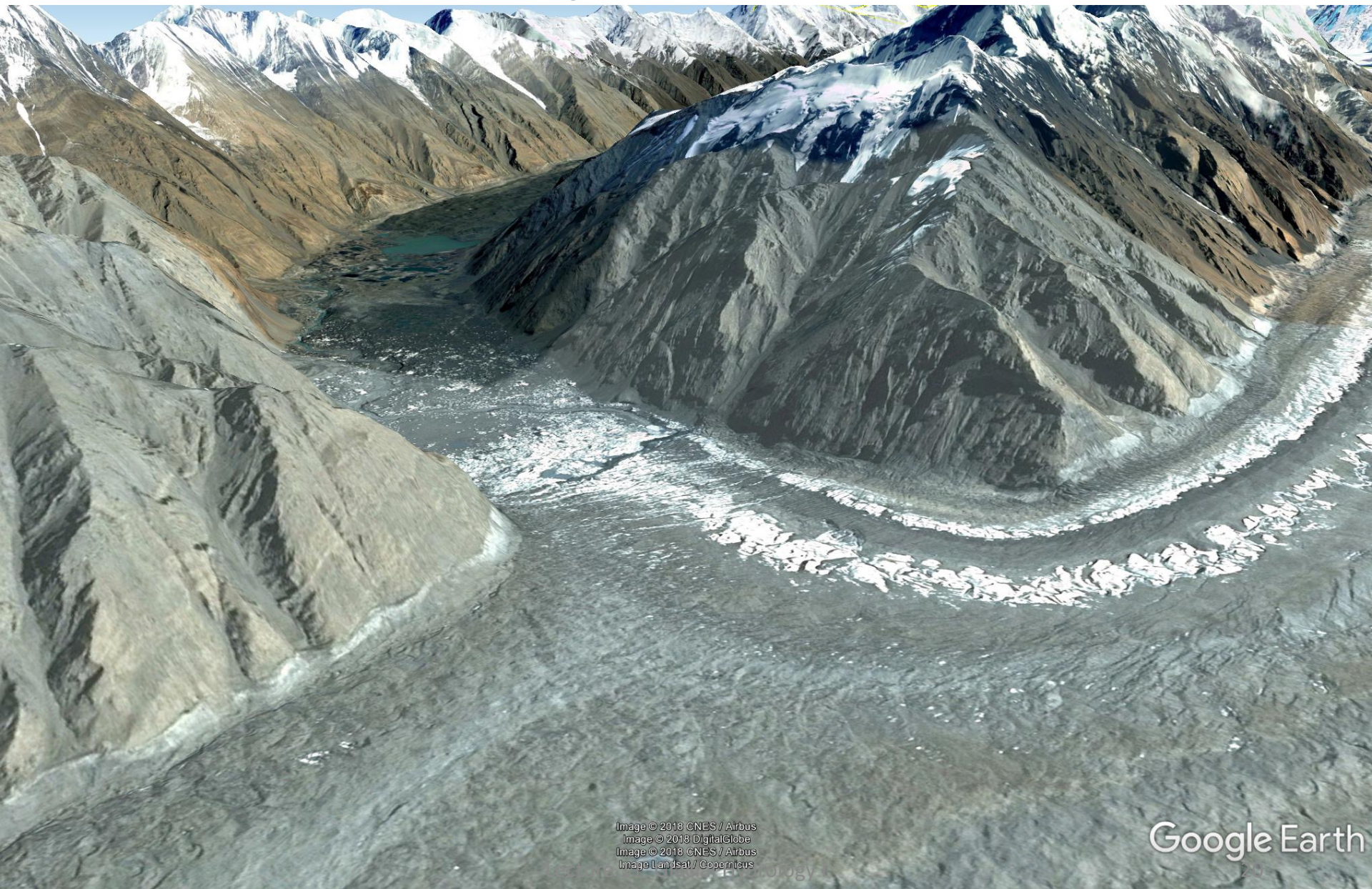
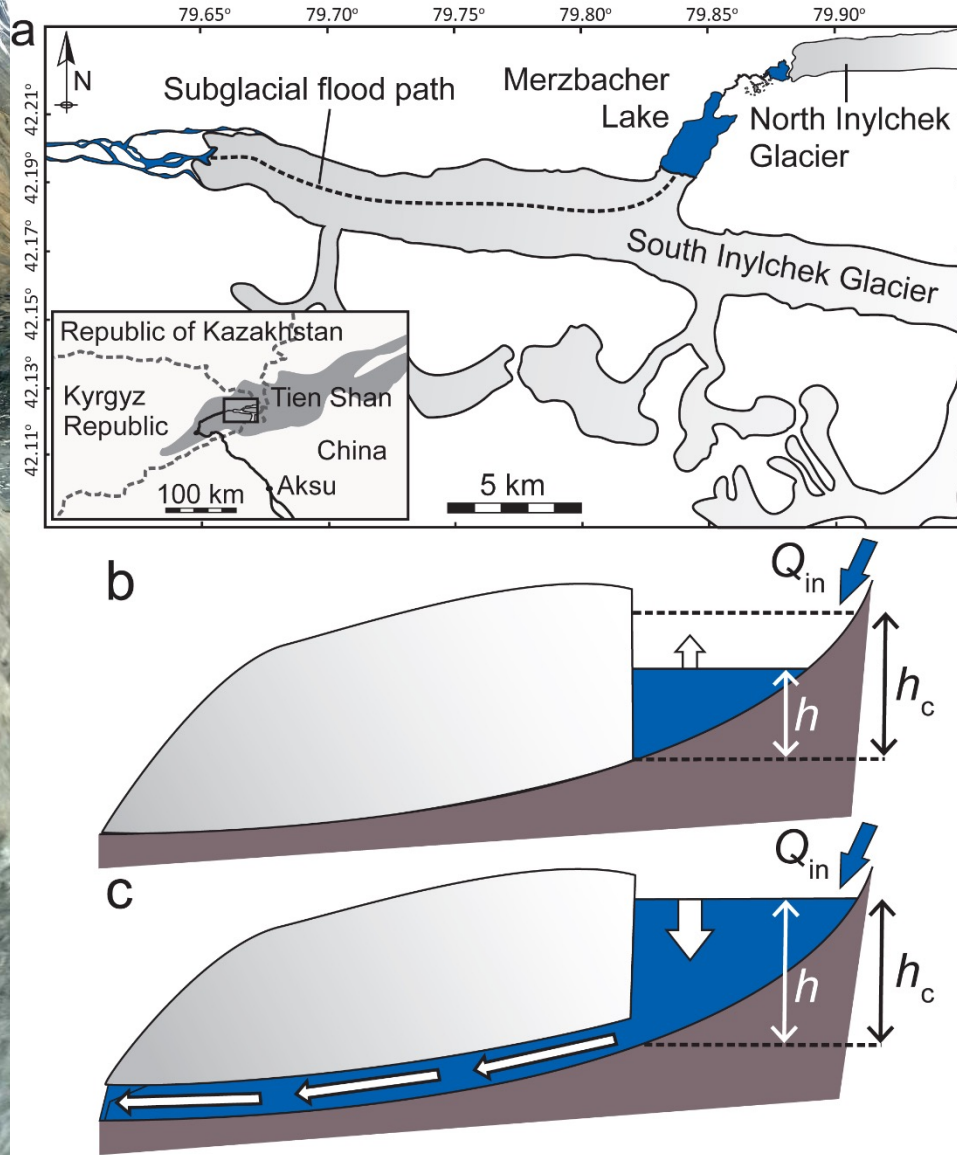


Image © 2018 CNES / Airbus
Image © 2018 DigitalGlobe
Image © 2018 CNES / Airbus
Image Landsat / Copernicus

Google Earth

Where? Mountain glaciers



Kingslake et al. (2013)

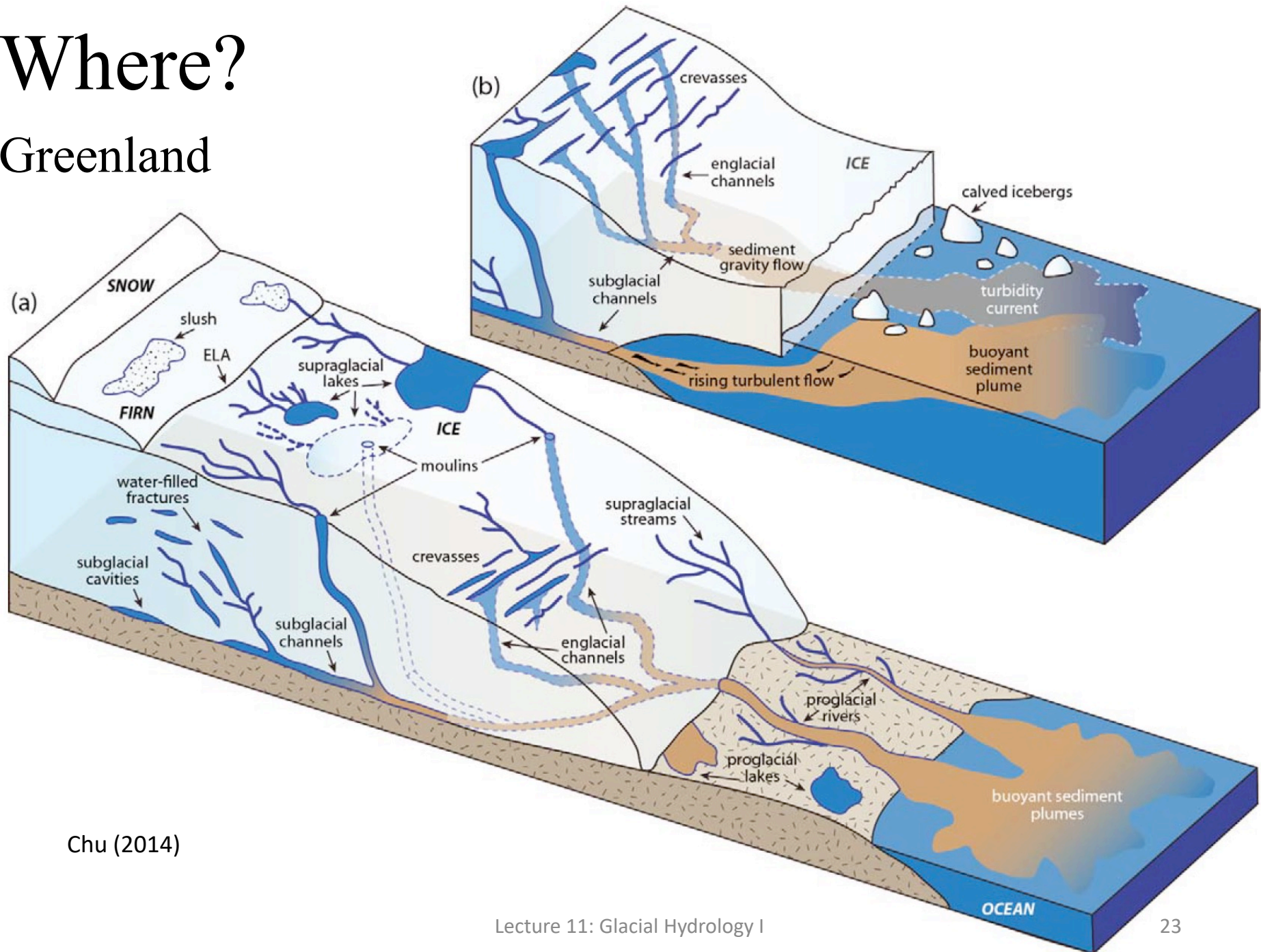
A satellite image of Greenland, showing the island's coastline and surrounding waters. The land is mostly covered in white snow and ice, with some green and brown patches indicating vegetation and bare ground. The surrounding waters are a deep blue. The word "Greenland" is written in a large, black, serif font across the center of the island.

Greenland

Image IBCAO
Image Landsat / Copernicus
Image U.S. Geological Survey
Data SIO, NOAA, U.S. Navy, NGA, GEBCO

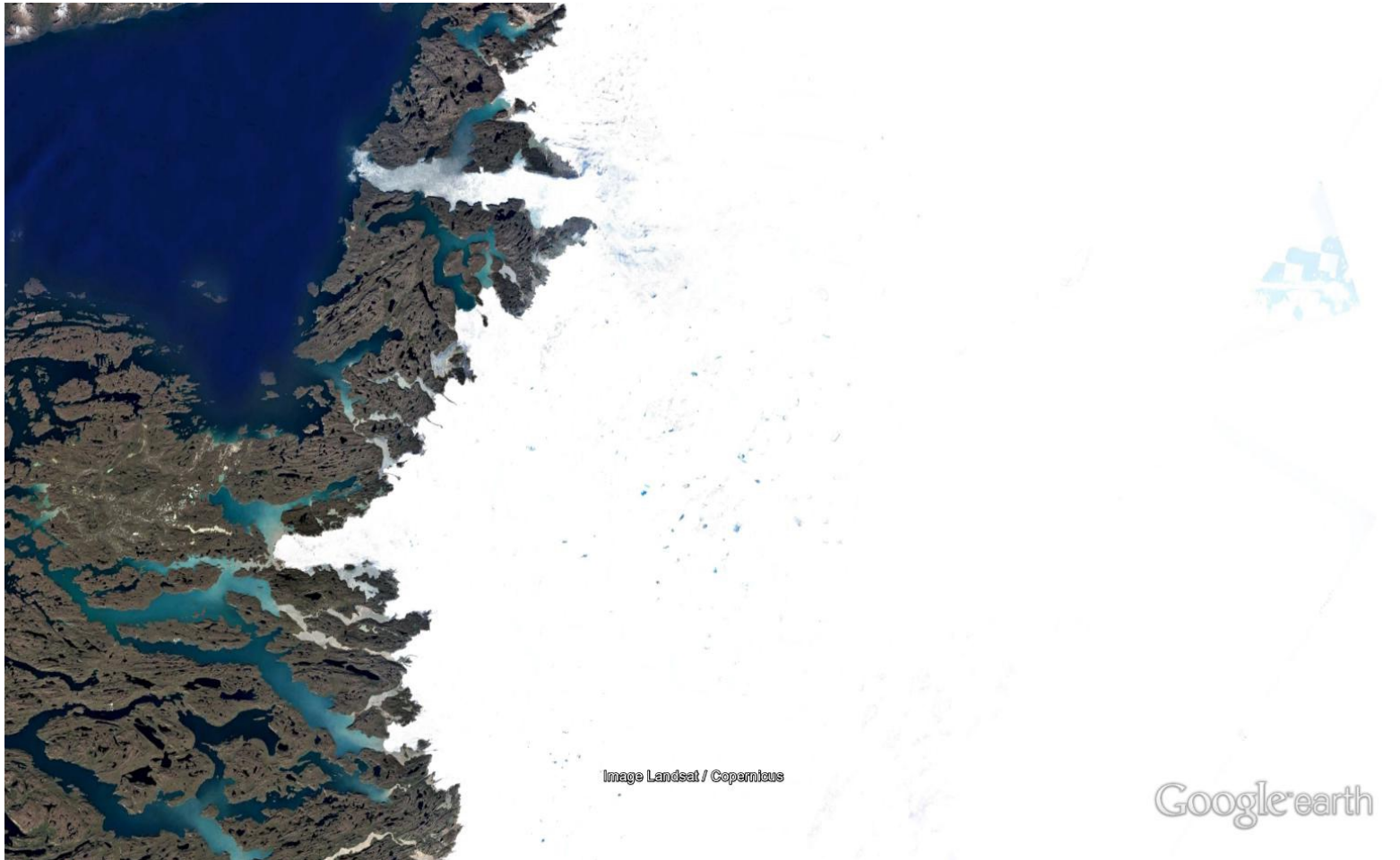
Where?

Greenland



Chu (2014)

Supraglacial Greenland



Supraglacial Greenland

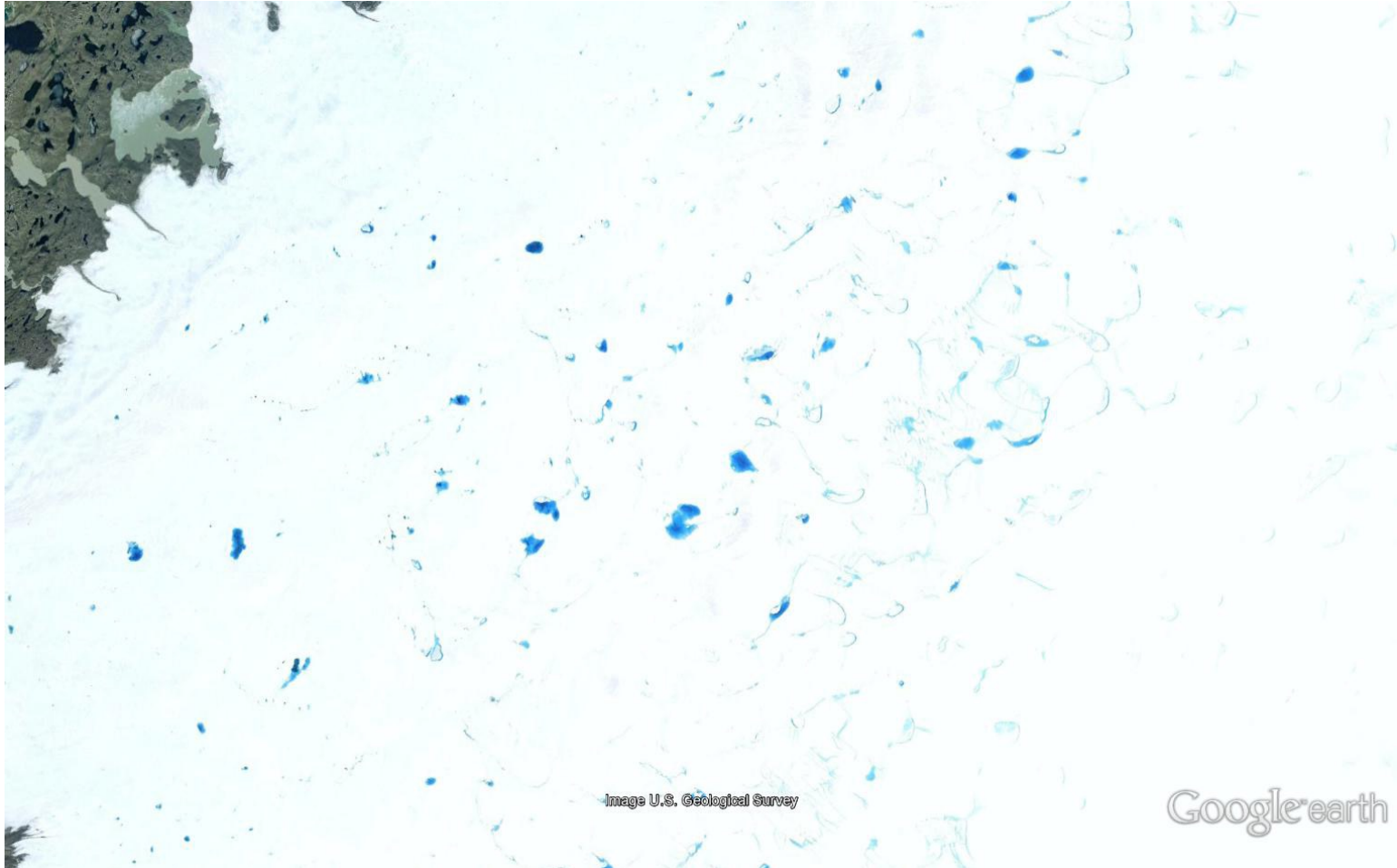
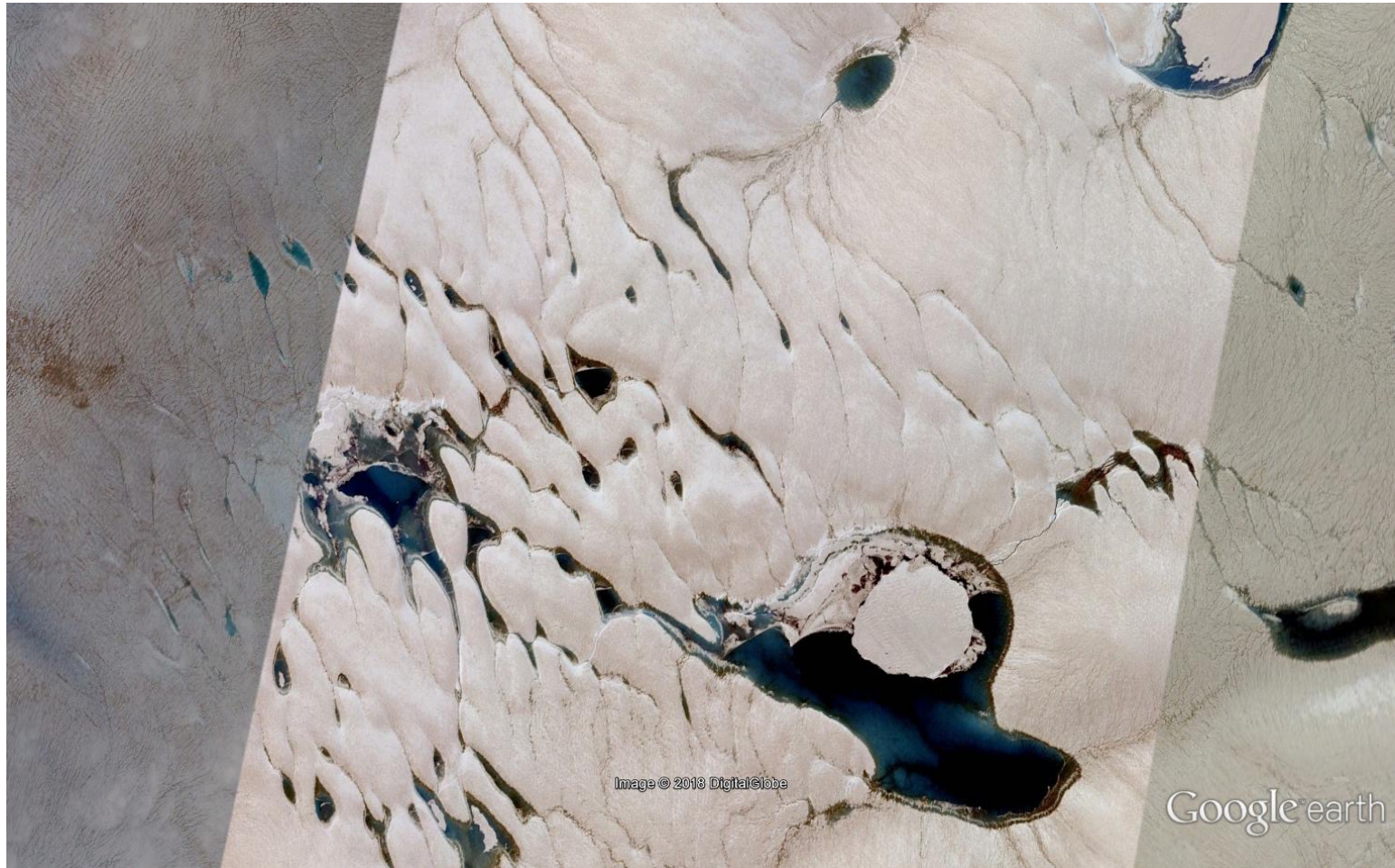


Image U.S. Geological Survey

Google earth

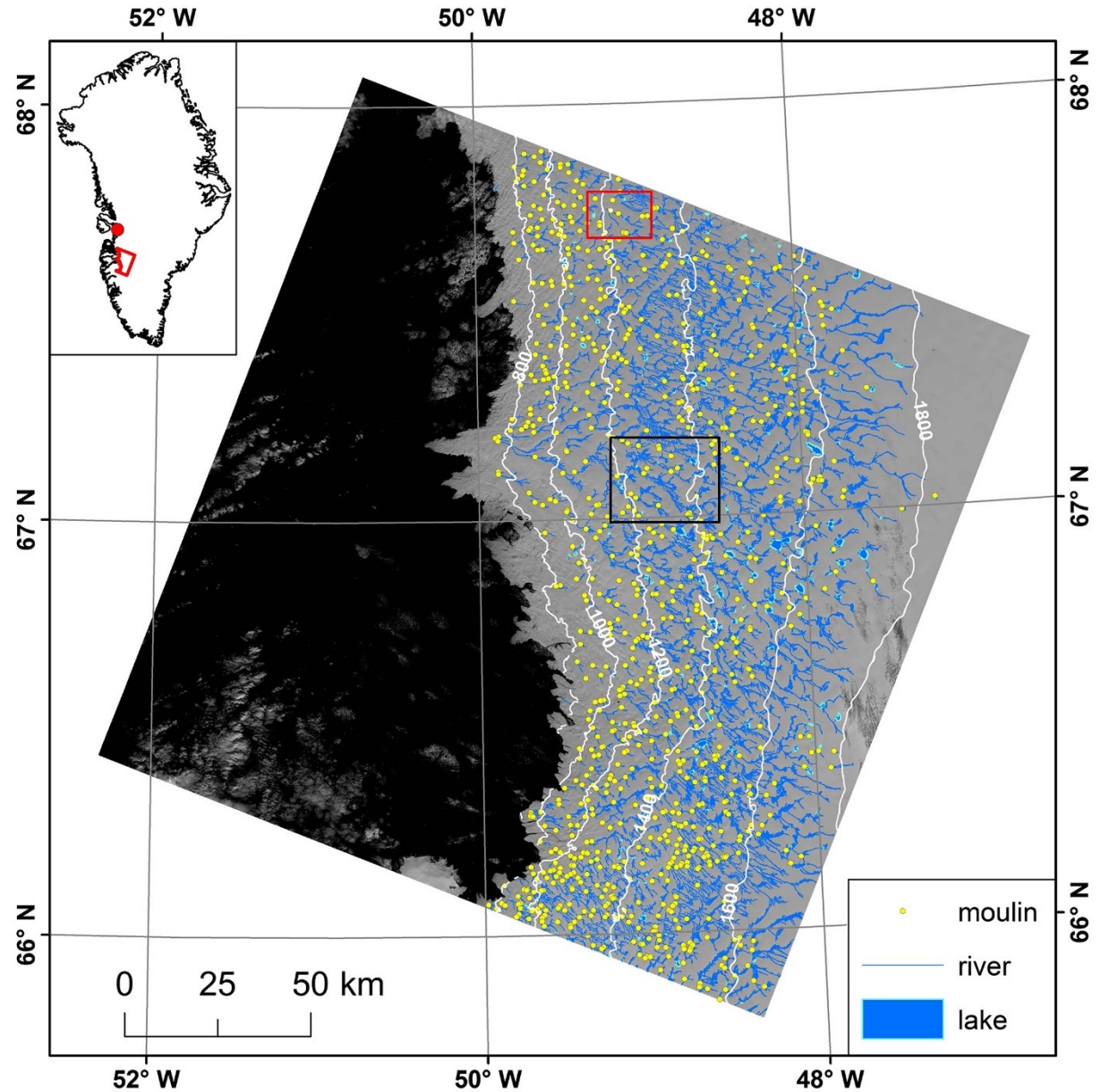
Supraglacial Greenland



Supraglacial Greenland

“Internally-drained
catchments” in South-
west Greenland

Yang and Smith (2016)



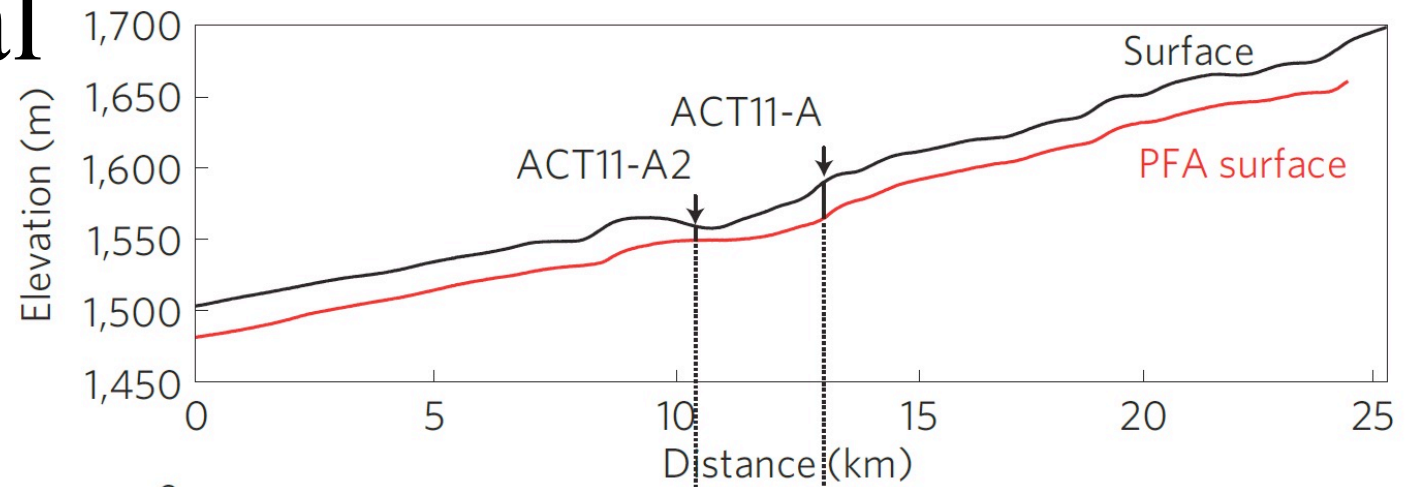
Englacial Greenland



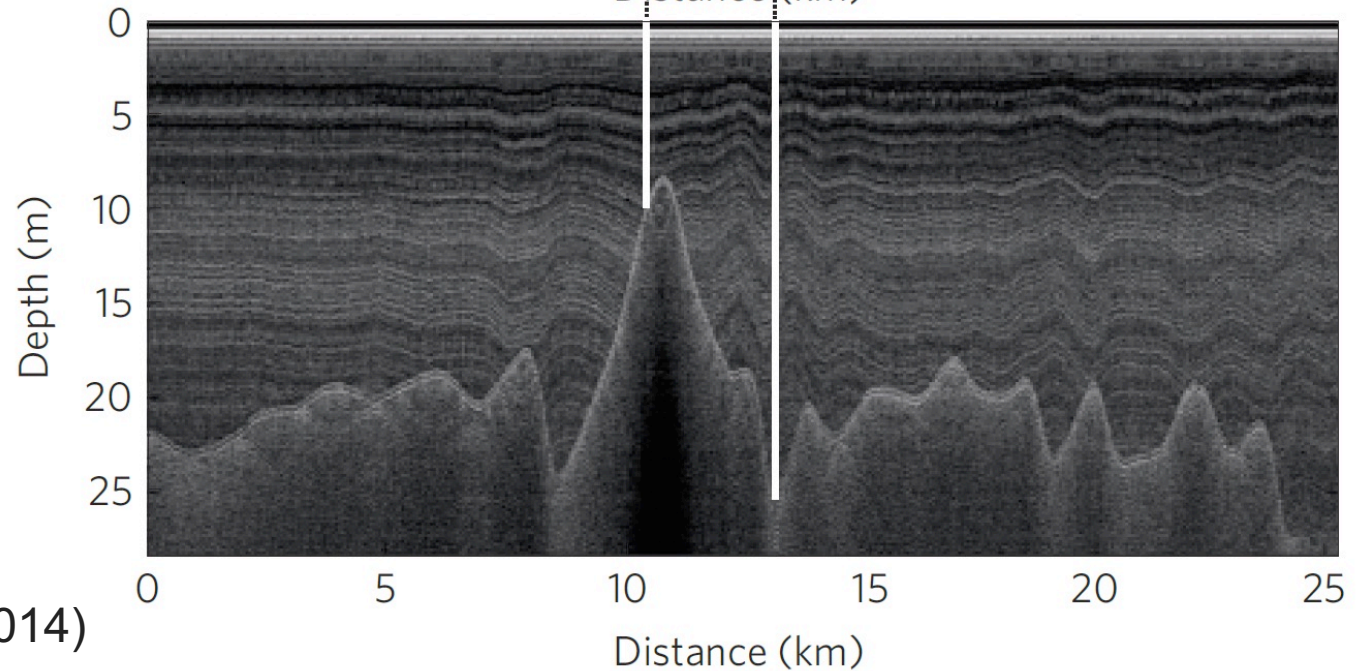
PHOTO COURTESY: ROGER BRAITHWAITE, UNIVERSITY OF MANCHESTER

**MELTING WATER FLOWING INTO A HUGE MOULIN
IN THE GREENLAND ICE SHEET**

Englacial Greenland



Radar can see the
top of a firn aquifer
in Greenland

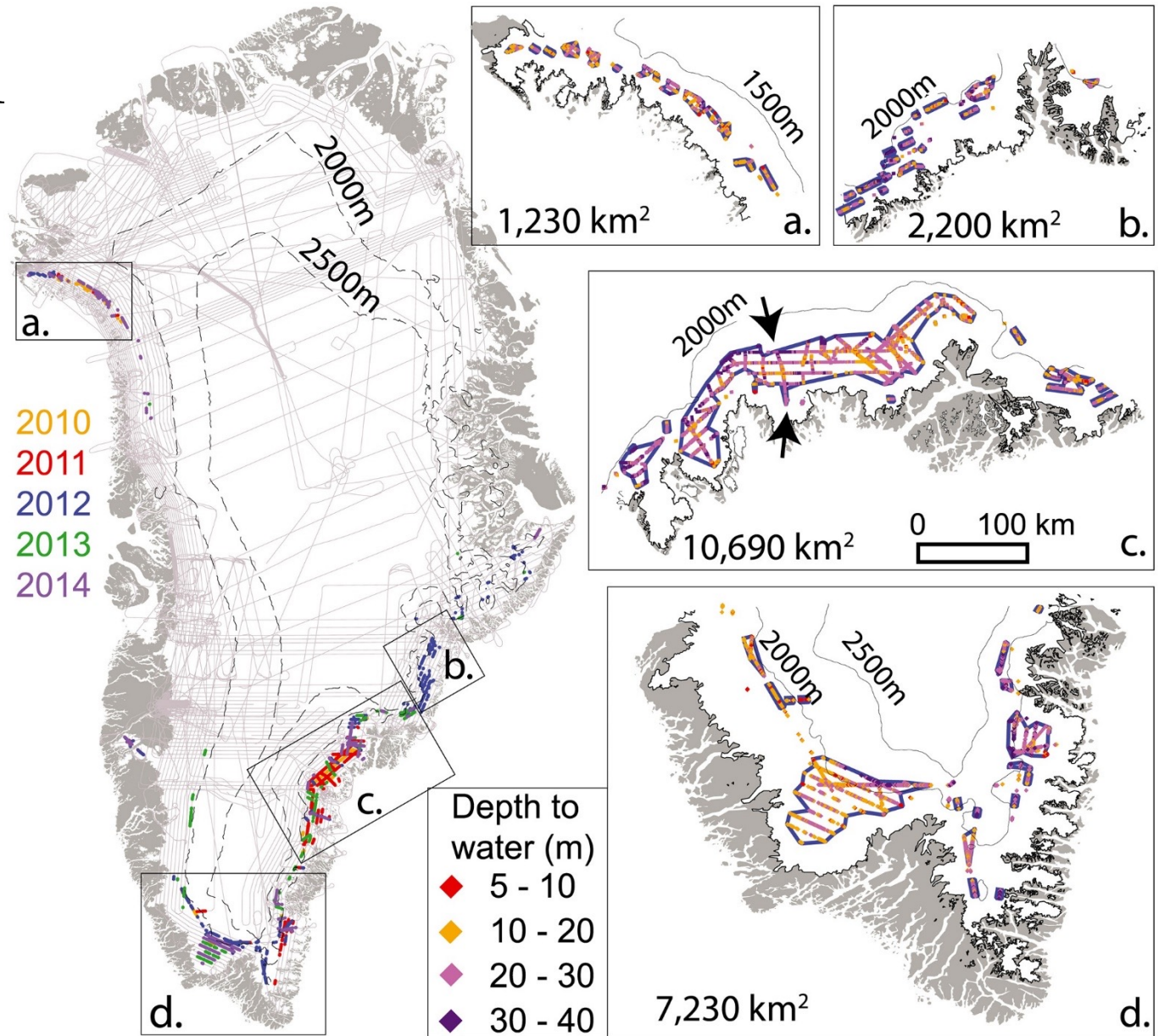


Forster et al. (2014)

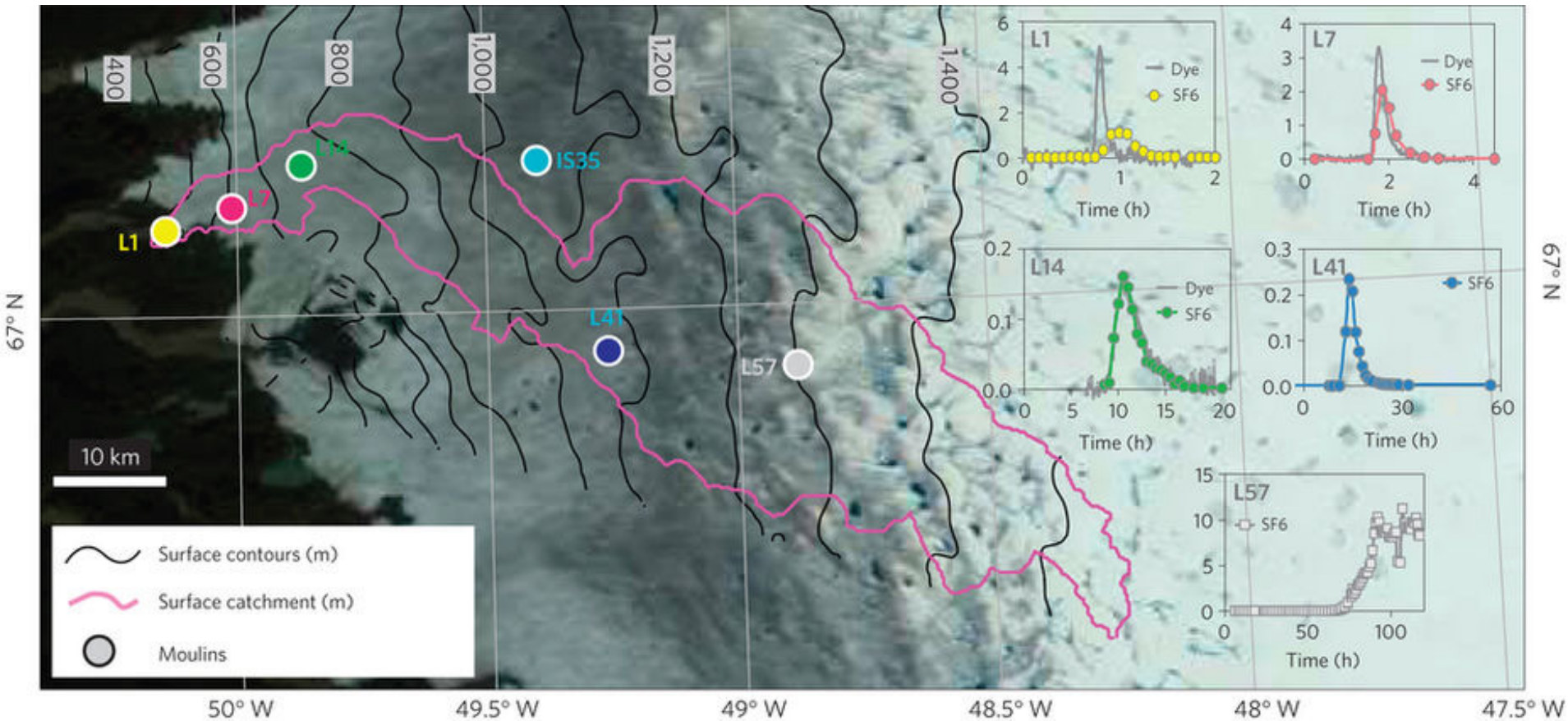
Englacial Greenland

Radar can see the
top of a firn aquifer
in Greenland

Miège et al. (2016)



Subglacial Greenland



Chandler et al. (2013)

Subglacial Greenland



Willis et al. (2014)

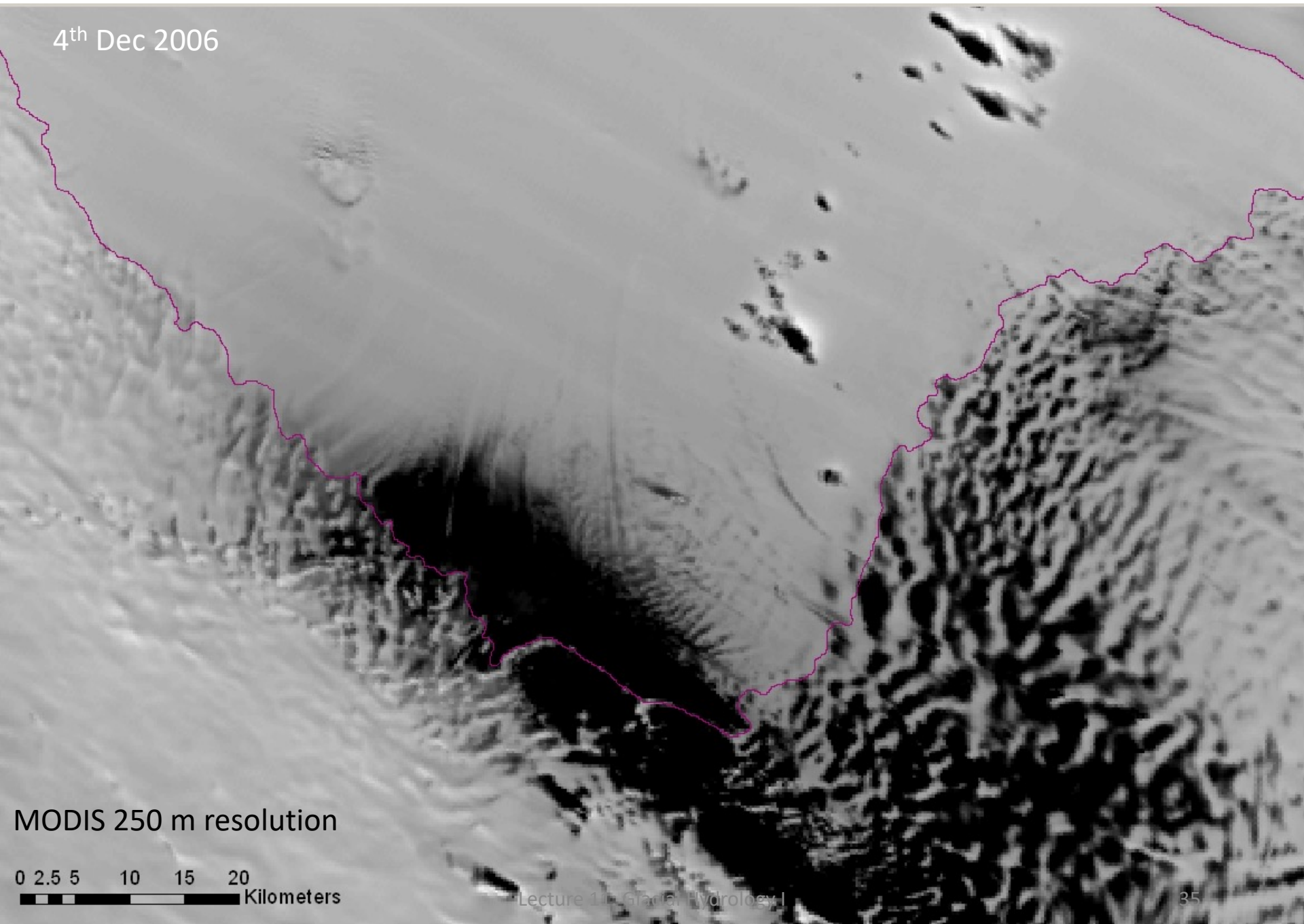
A satellite image of the continent of Antarctica, showing its white ice surface and the surrounding blue ocean. The continent is centered in the frame, with the word "Antarctica" overlaid in the middle.

Antarctica

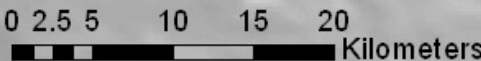
Image U.S. Geological Survey
Image Landsat / Copernicus
Image PGC/NASA
Figure 11. Glacial Hydrology I
Data SIO, NOAA, U.S. Navy, NGA, GEBCO

Nivlisen Ice Shelf, East Antarctica 2006 – 2007 summer

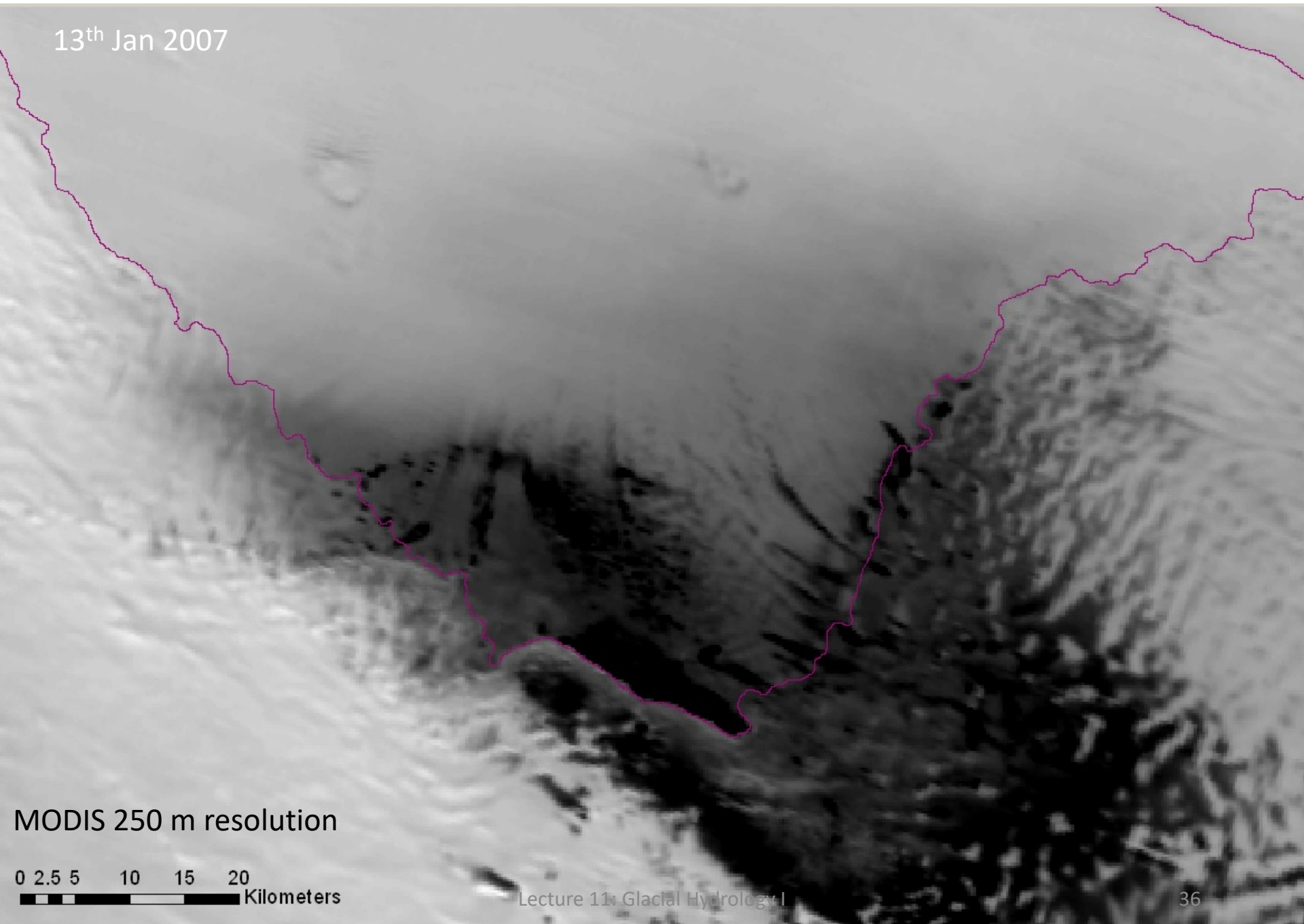
4th Dec 2006



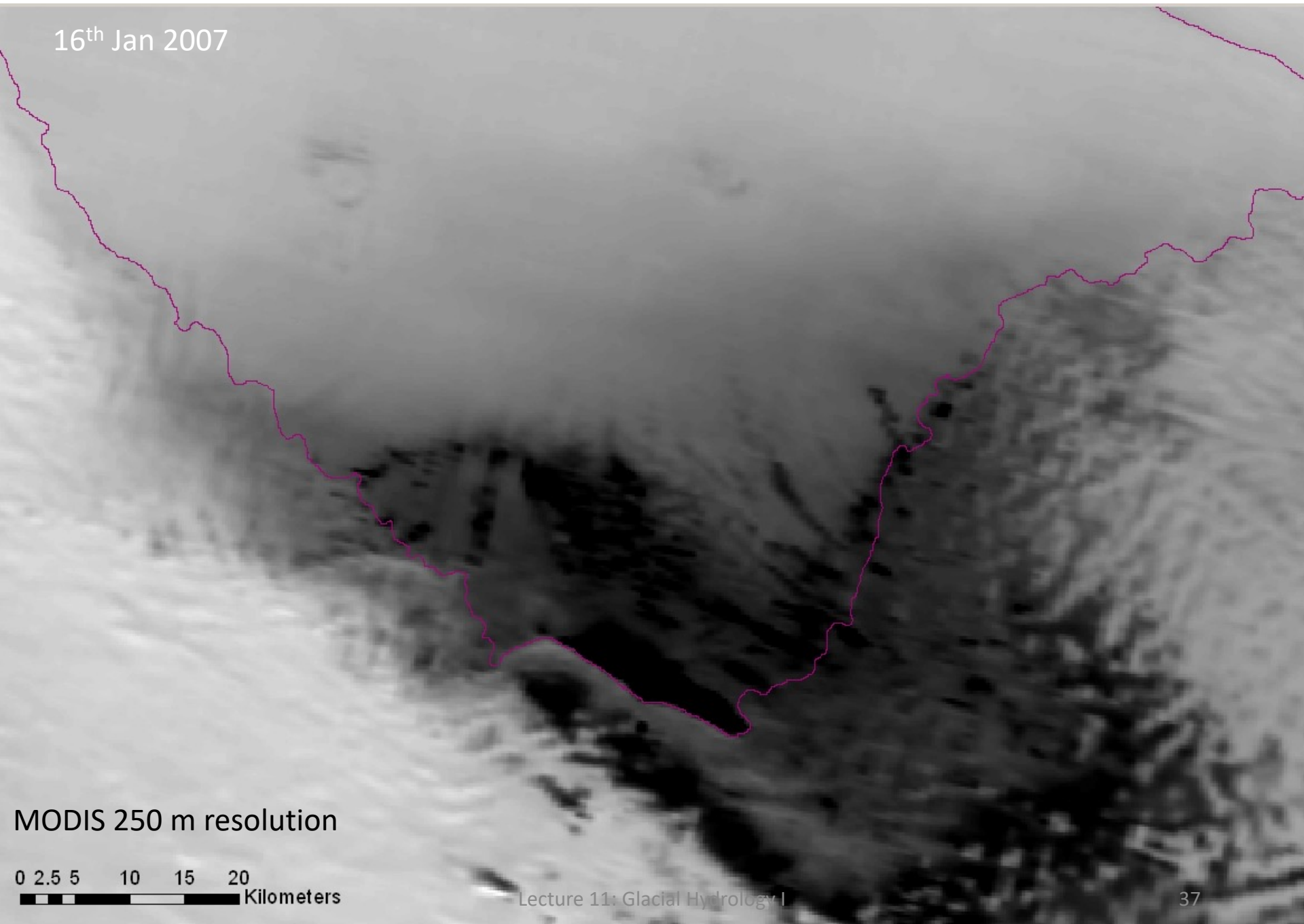
MODIS 250 m resolution



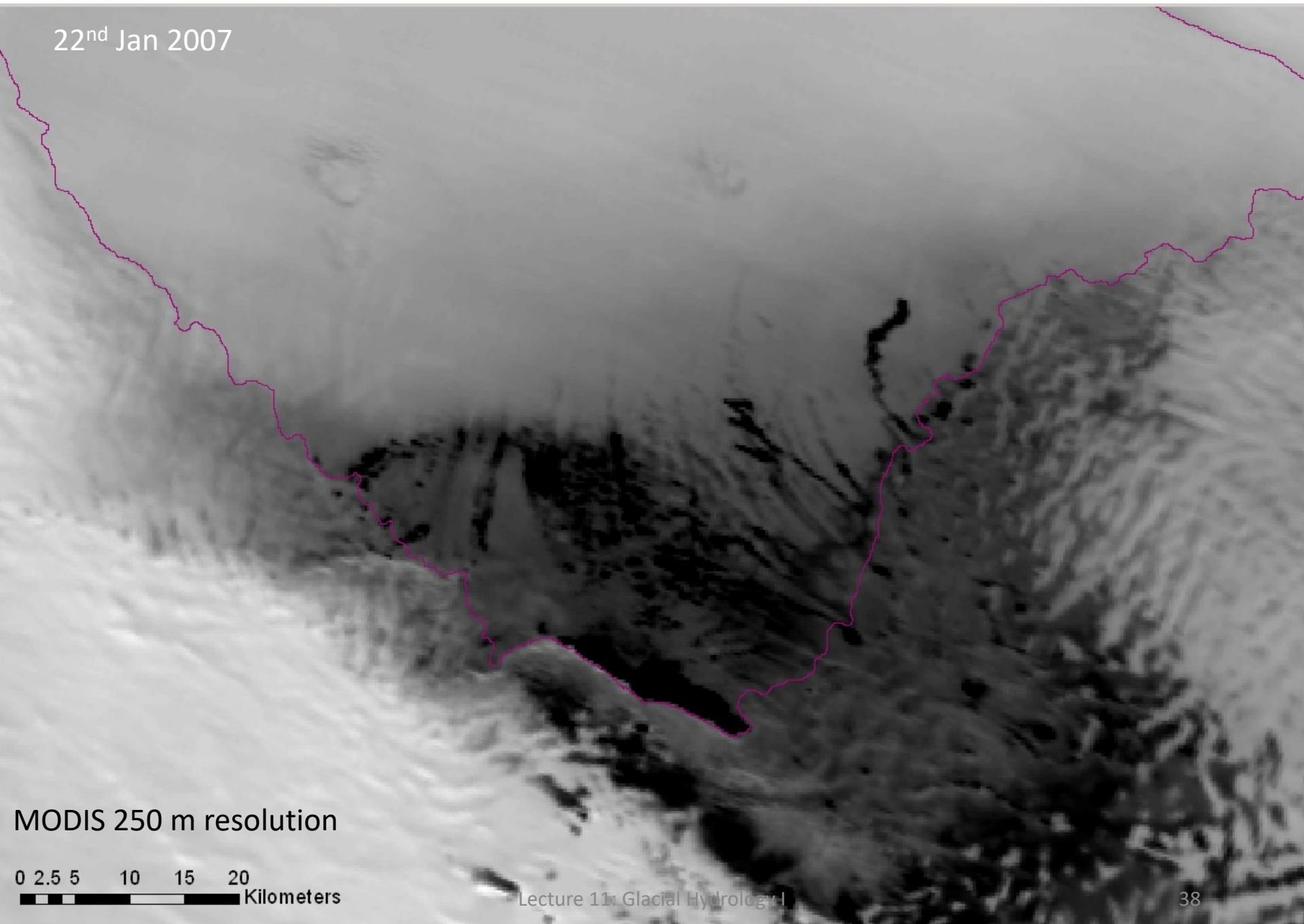
13th Jan 2007



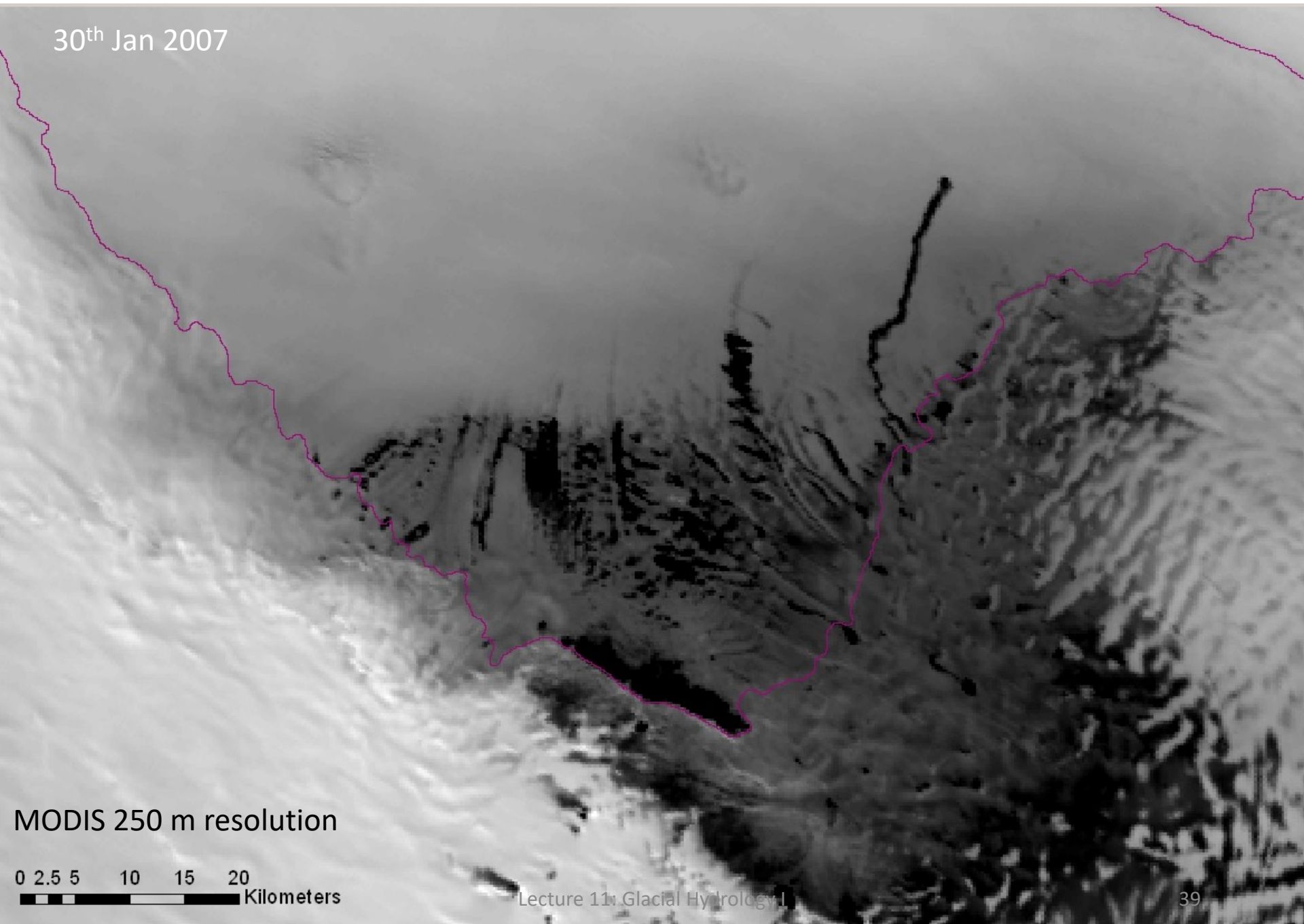
16th Jan 2007



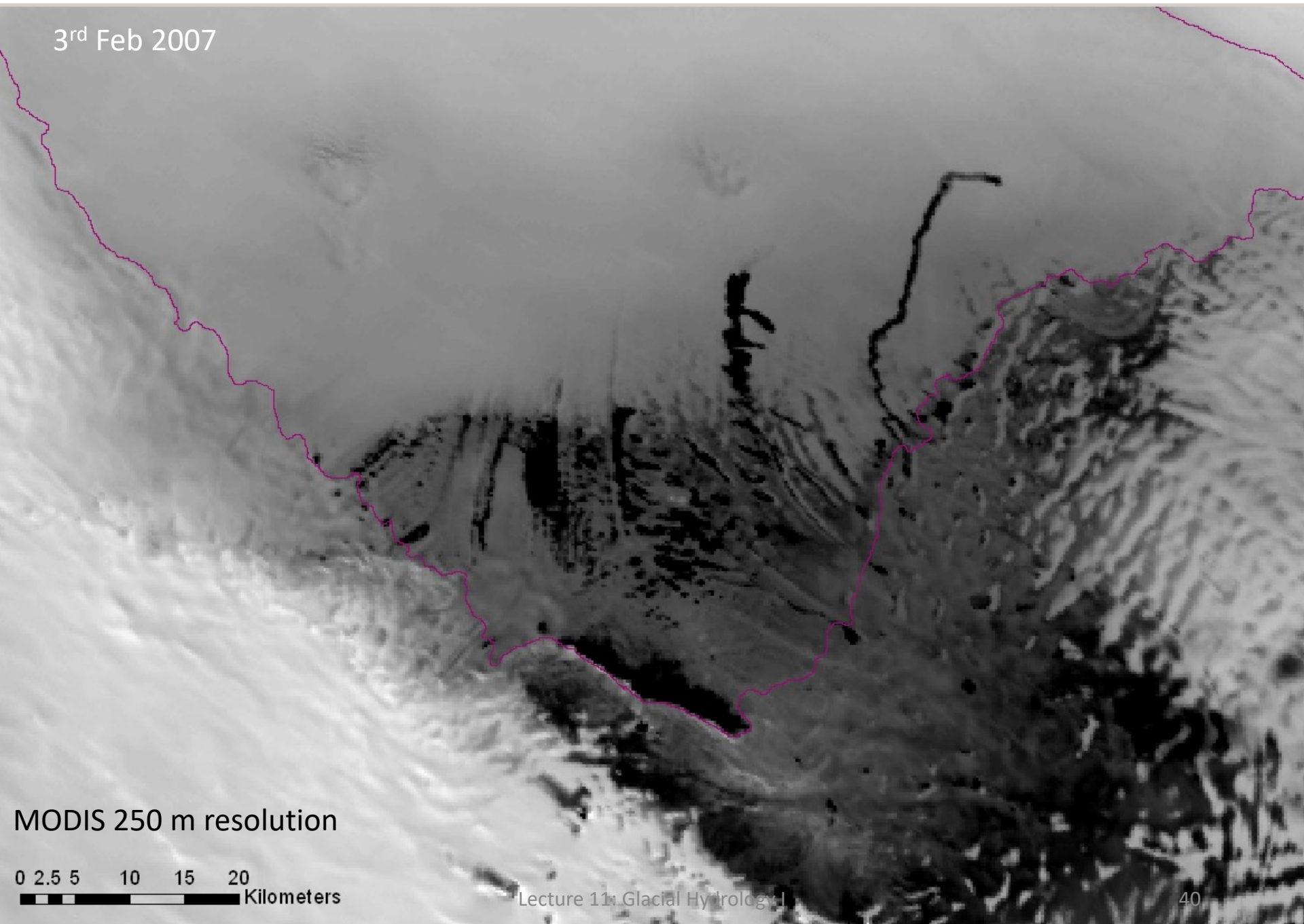
22nd Jan 2007



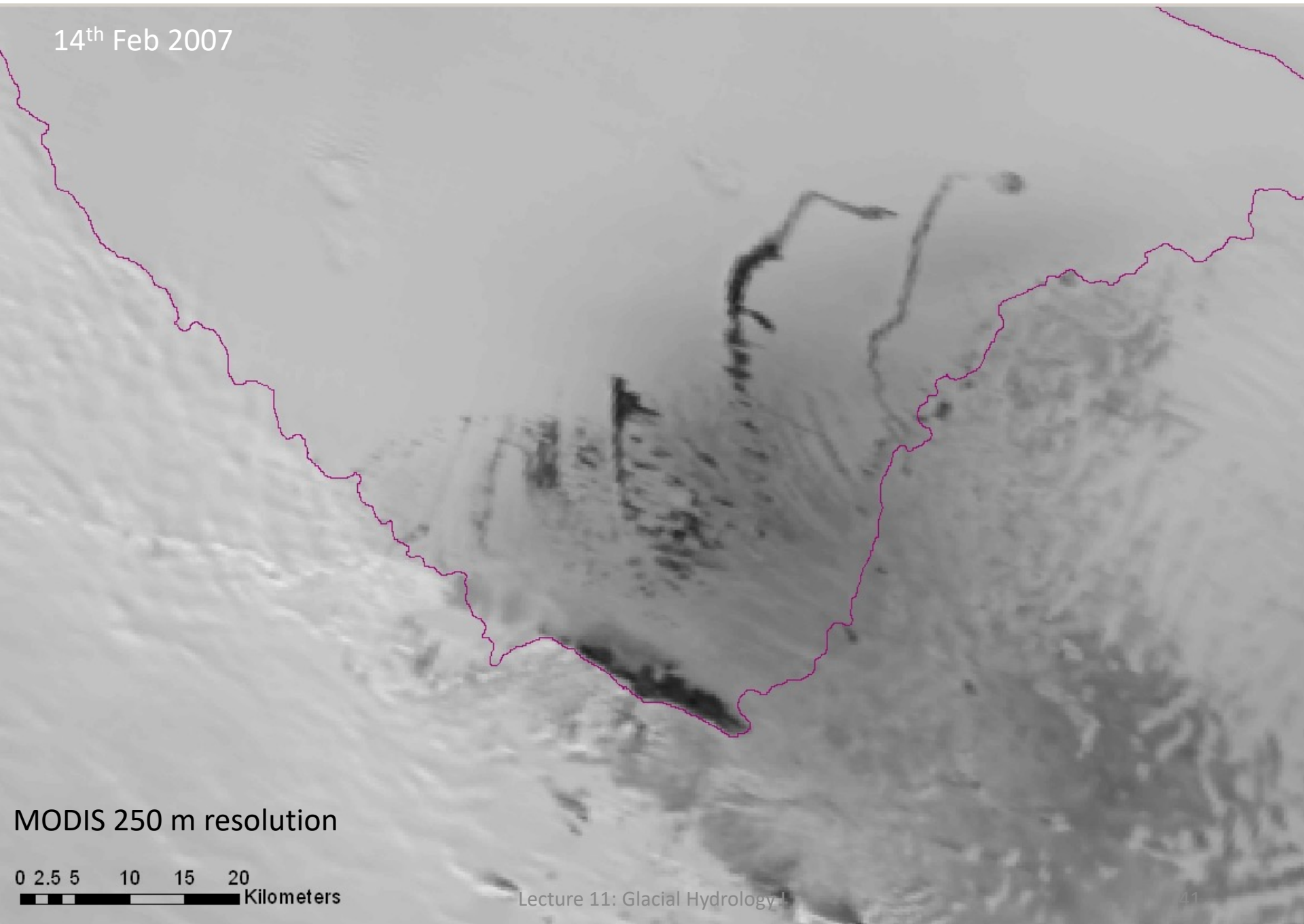
30th Jan 2007



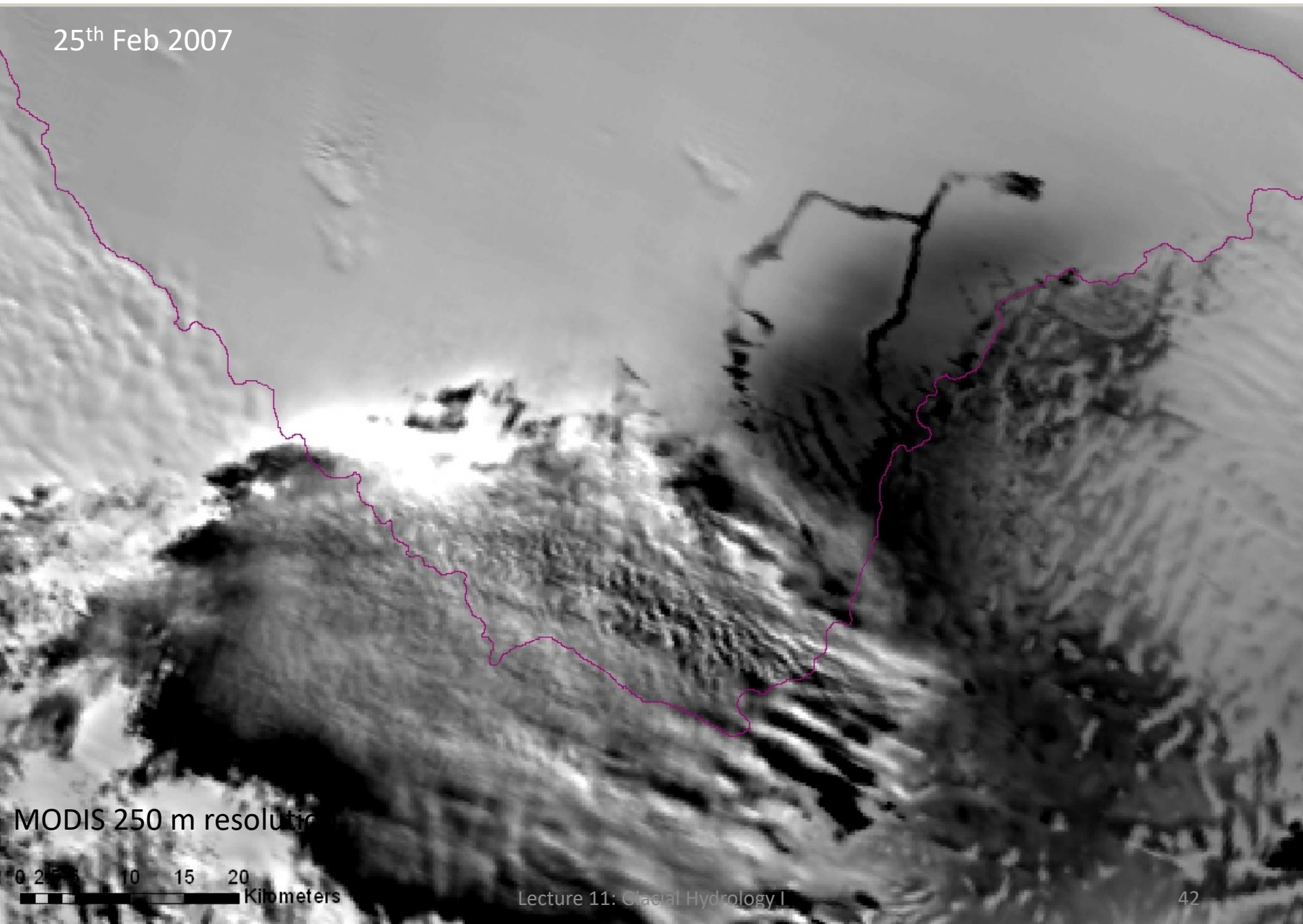
3rd Feb 2007



14th Feb 2007



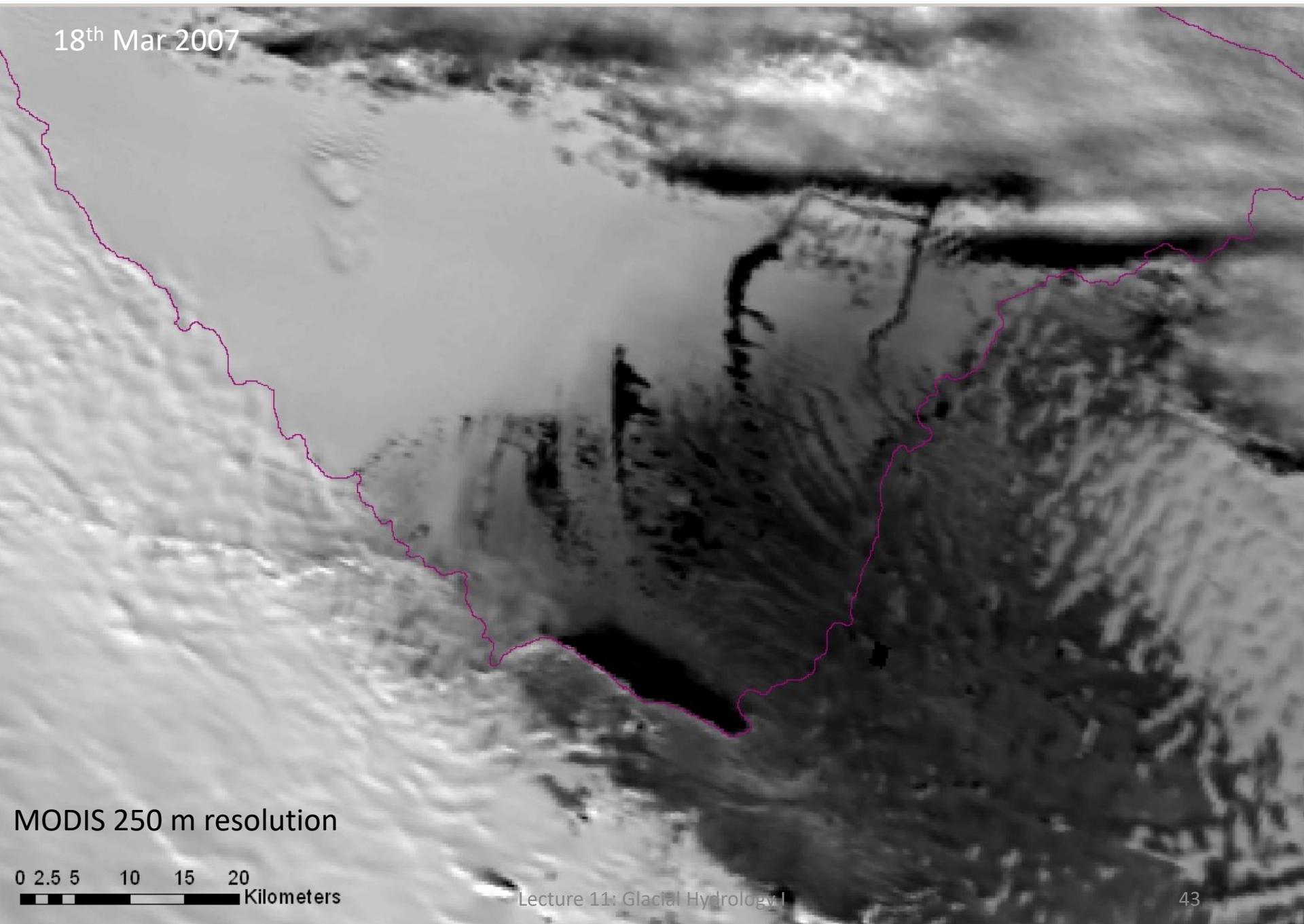
25th Feb 2007



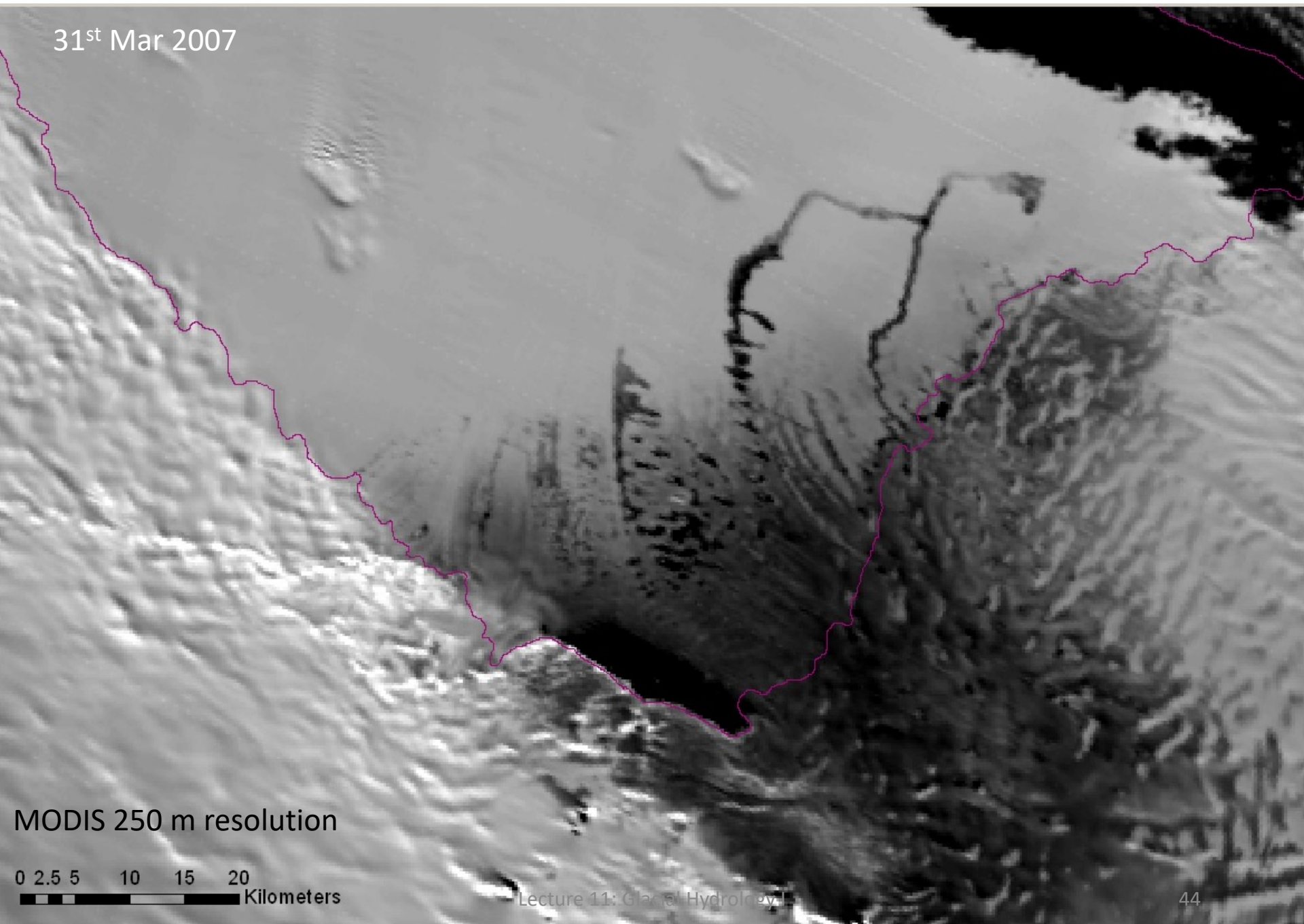
MODIS 250 m resolution

0 2.5 10 15 20
Kilometers

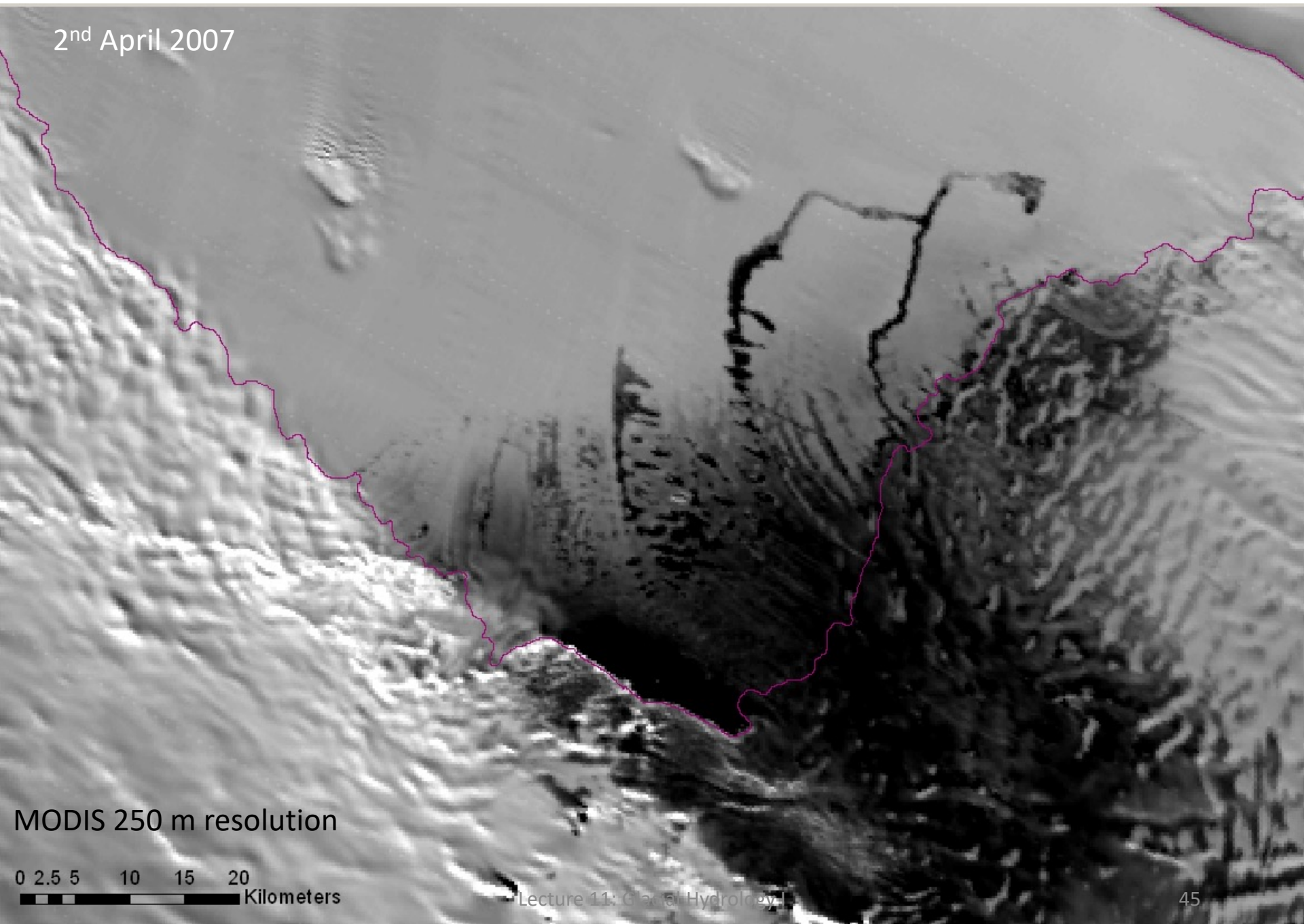
18th Mar 2007



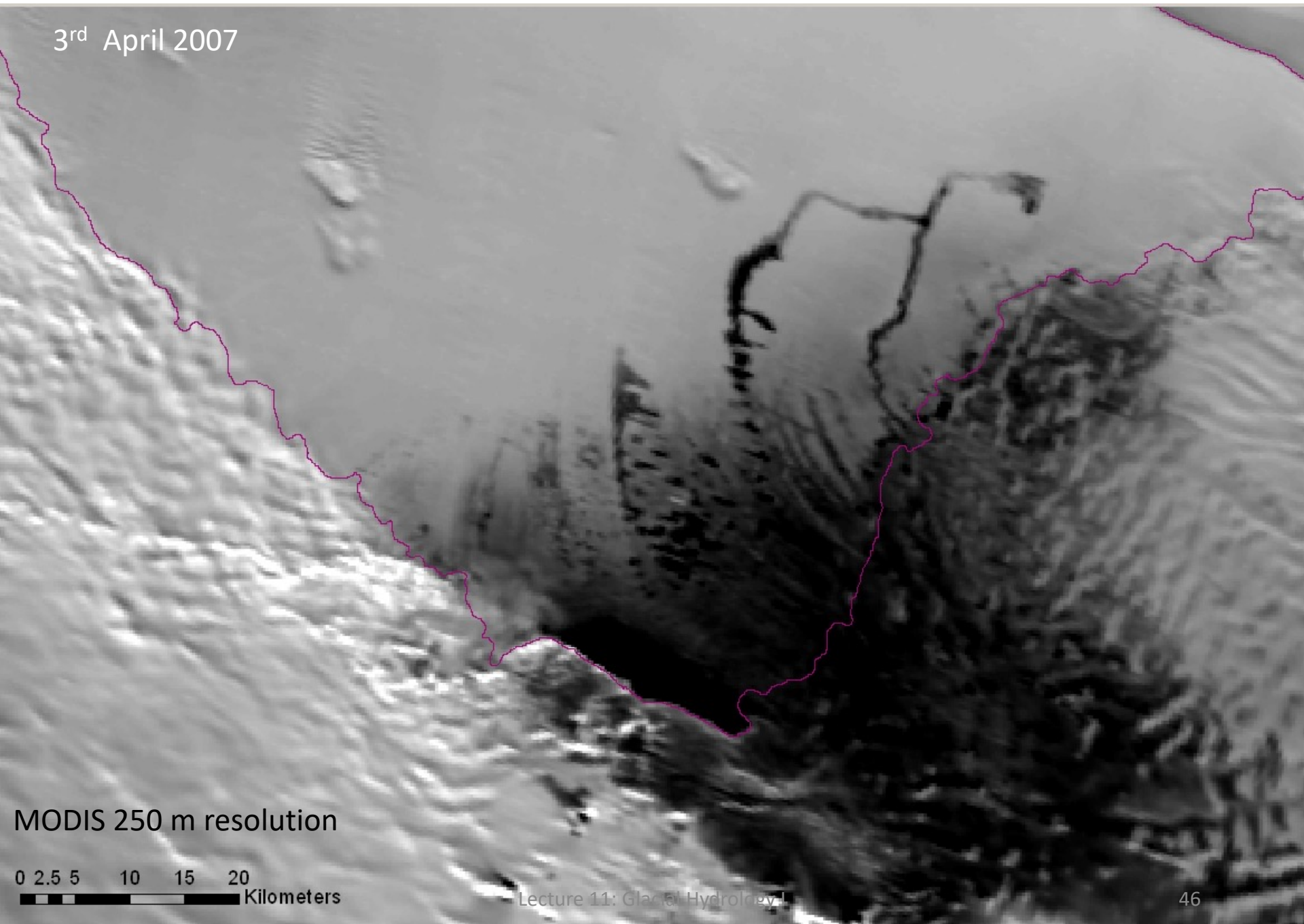
31st Mar 2007



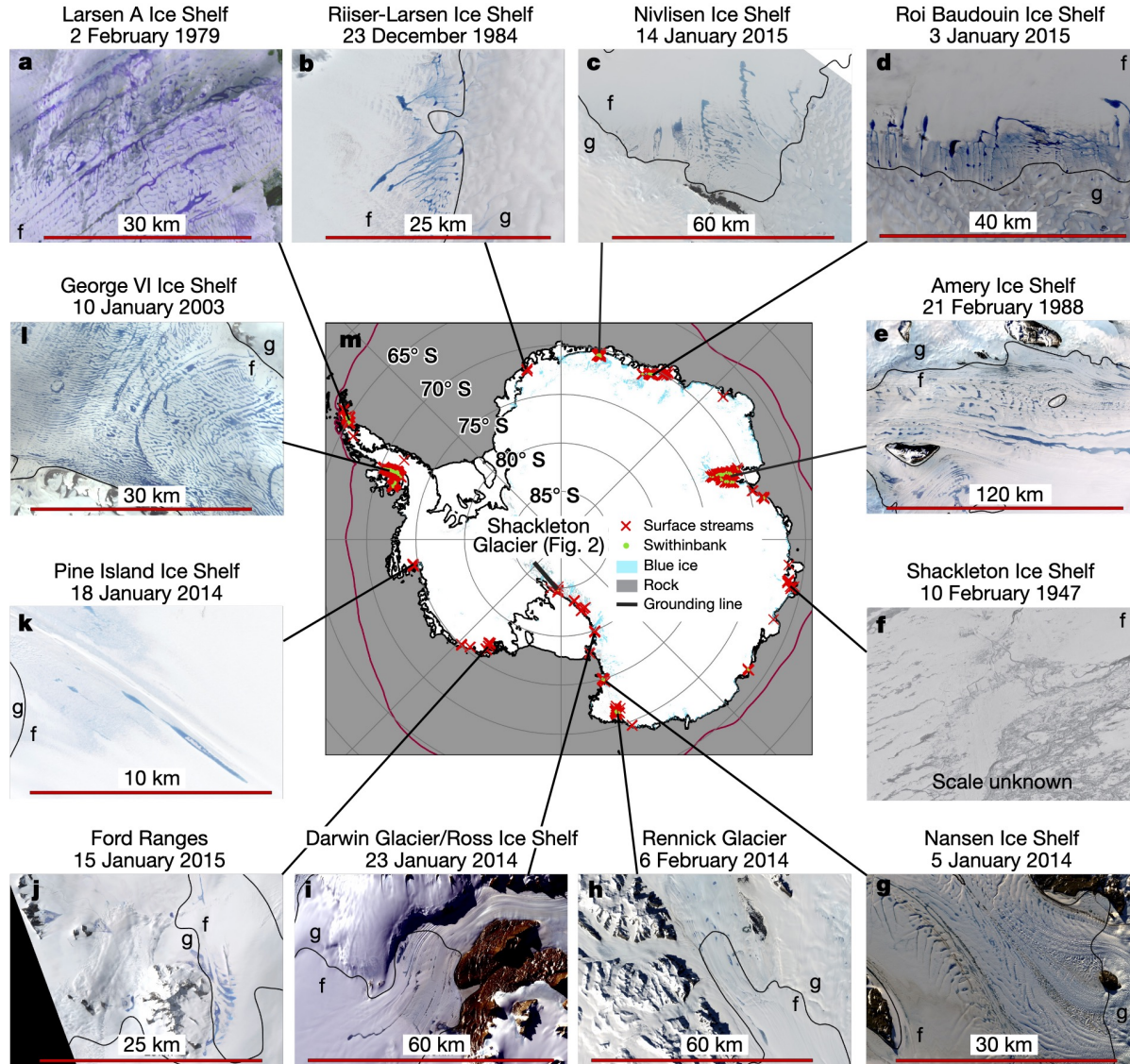
2nd April 2007



3rd April 2007



Supraglacial Antarctica

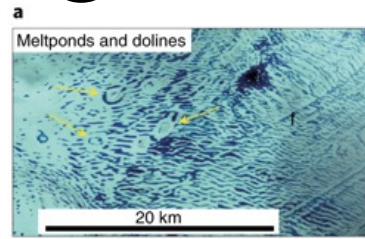


Kingslake et al. (2017)

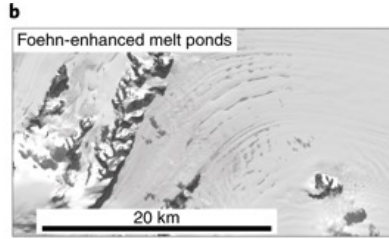
Lecture 11: Glacial Hydrology I

Supraglacial

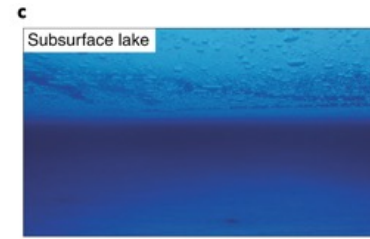
Antarctica



George VI Ice Shelf
10 January 2003



Larsen C Ice Shelf
7 February 2016



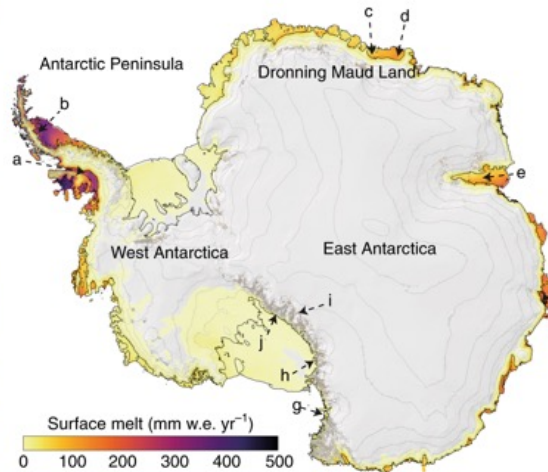
Western Roi Baudouin Ice Shelf
January/February 2016



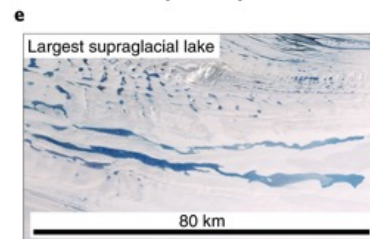
Shackleton Glacier
10 January 2010



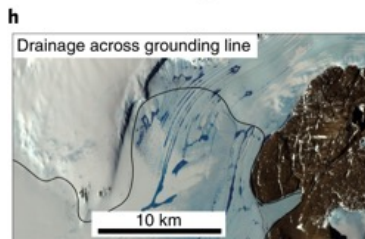
Law Glacier (~1,830 m a.s.l.)
14 January 2017



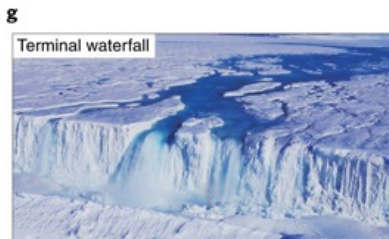
Eastern Roi Baudouin Ice Shelf
January/February 2016



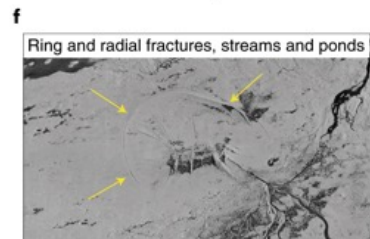
Amery Ice Shelf
21 February 1988



Darwin Glacier/Ross Ice Shelf
31 December 2001



Nansen Ice Shelf
12 January 2014



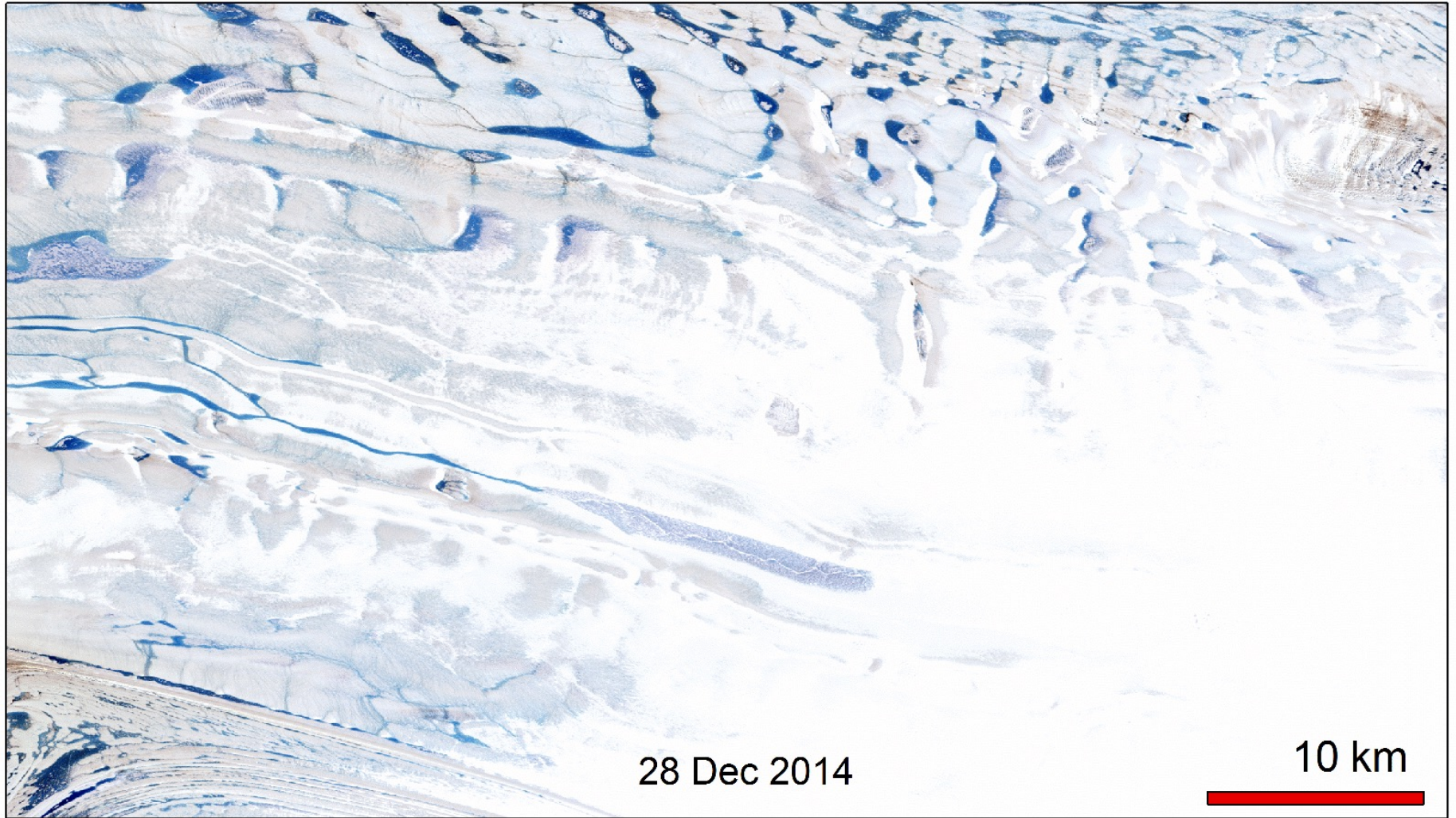
Shackleton Ice Shelf
10 February 1947

Bell et al. (2018)

Lecture 11: Glacial Hydrology I

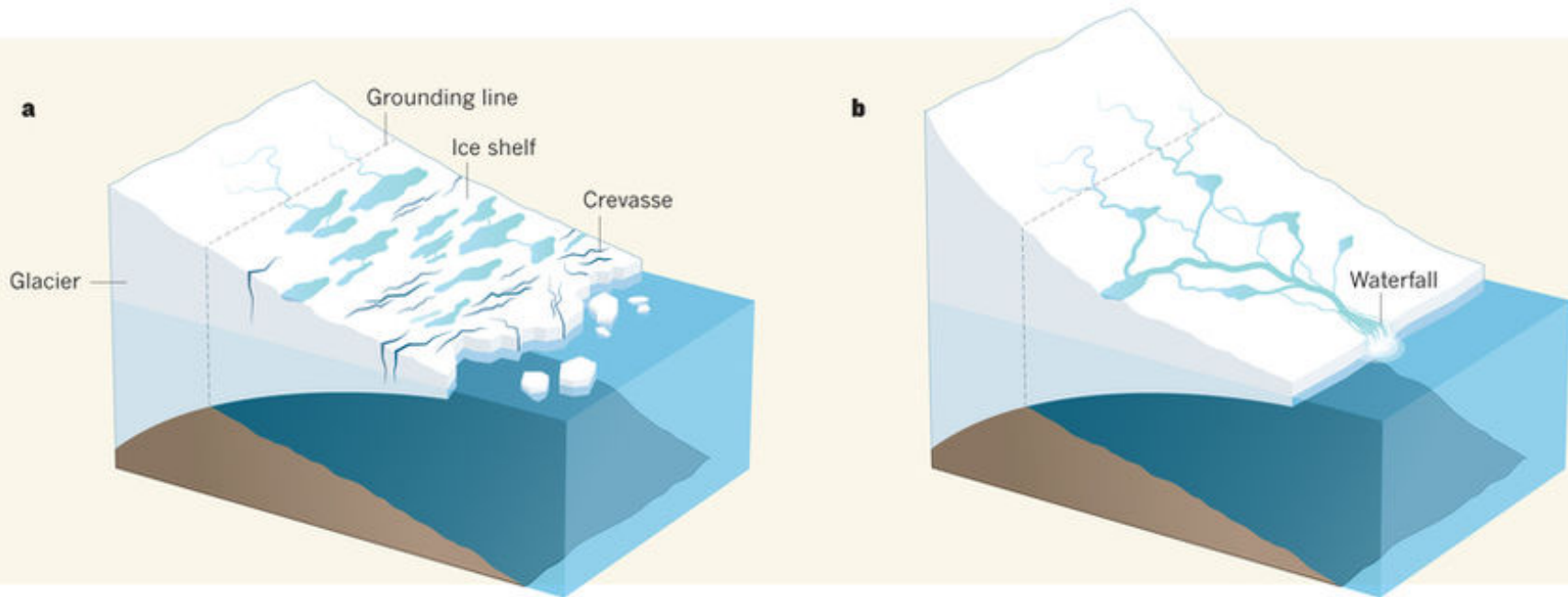
Supraglacial

Amery Ice Shelf



Spergel et al., 2020

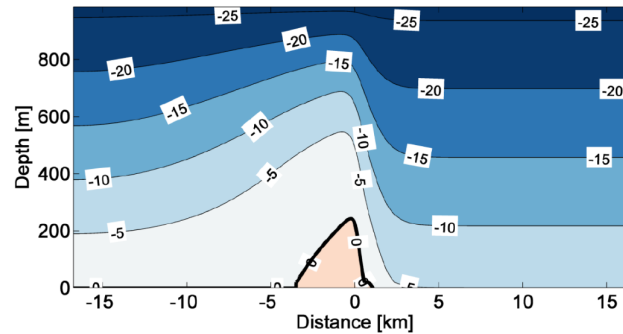
Supraglacial Antarctica



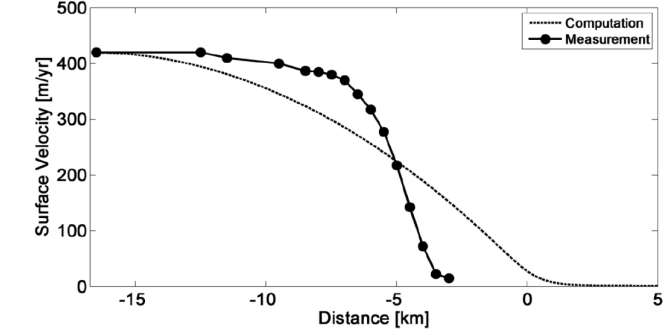
Banwell (2014)

Englacial Antarctica

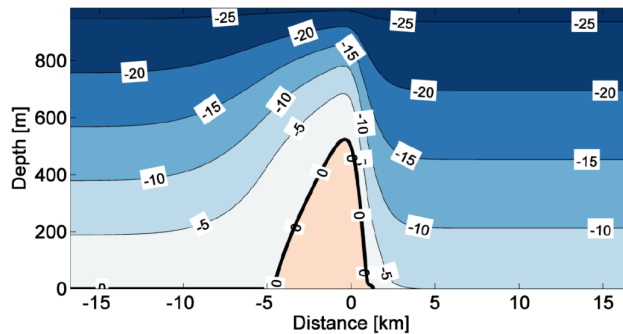
A1. Temperature (linear rheology)



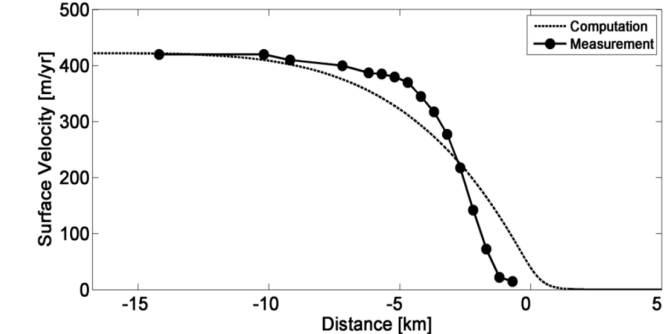
B1. Surface velocities (linear rheology)



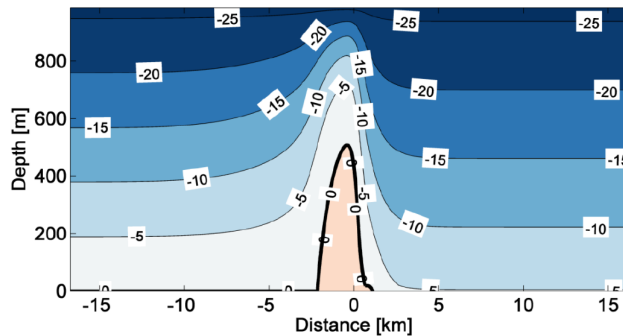
A2. Temperature (nonlinear, temperature-independent rheology)



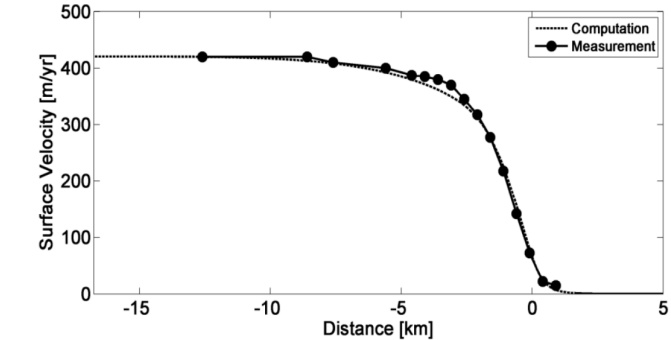
B2. Surface velocities (nonlinear, temperature-independent rheology)



A3. Temperature (nonlinear, temperature-dependent rheology)

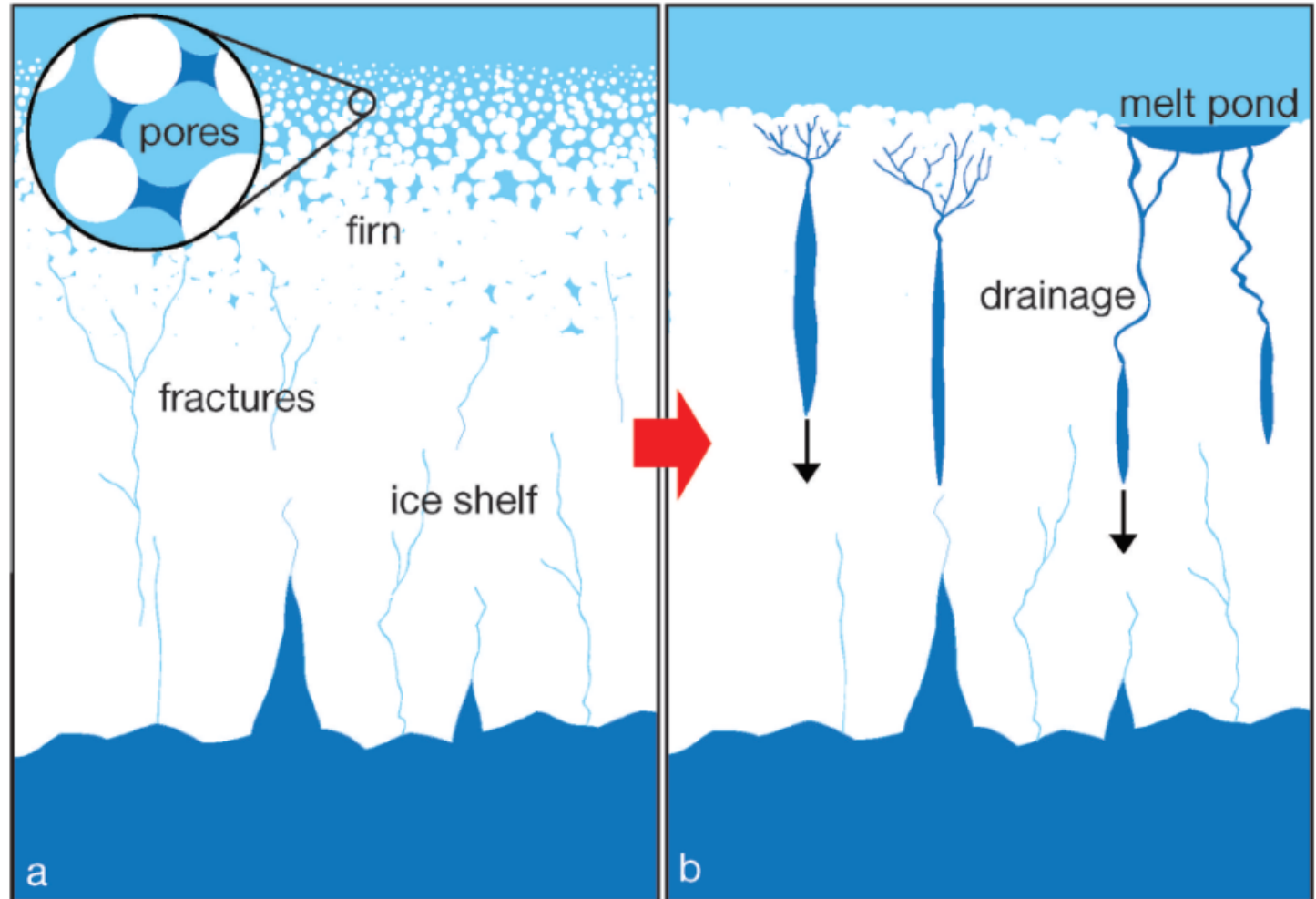


B3. Surface velocities (nonlinear, temperature-dependent rheology)



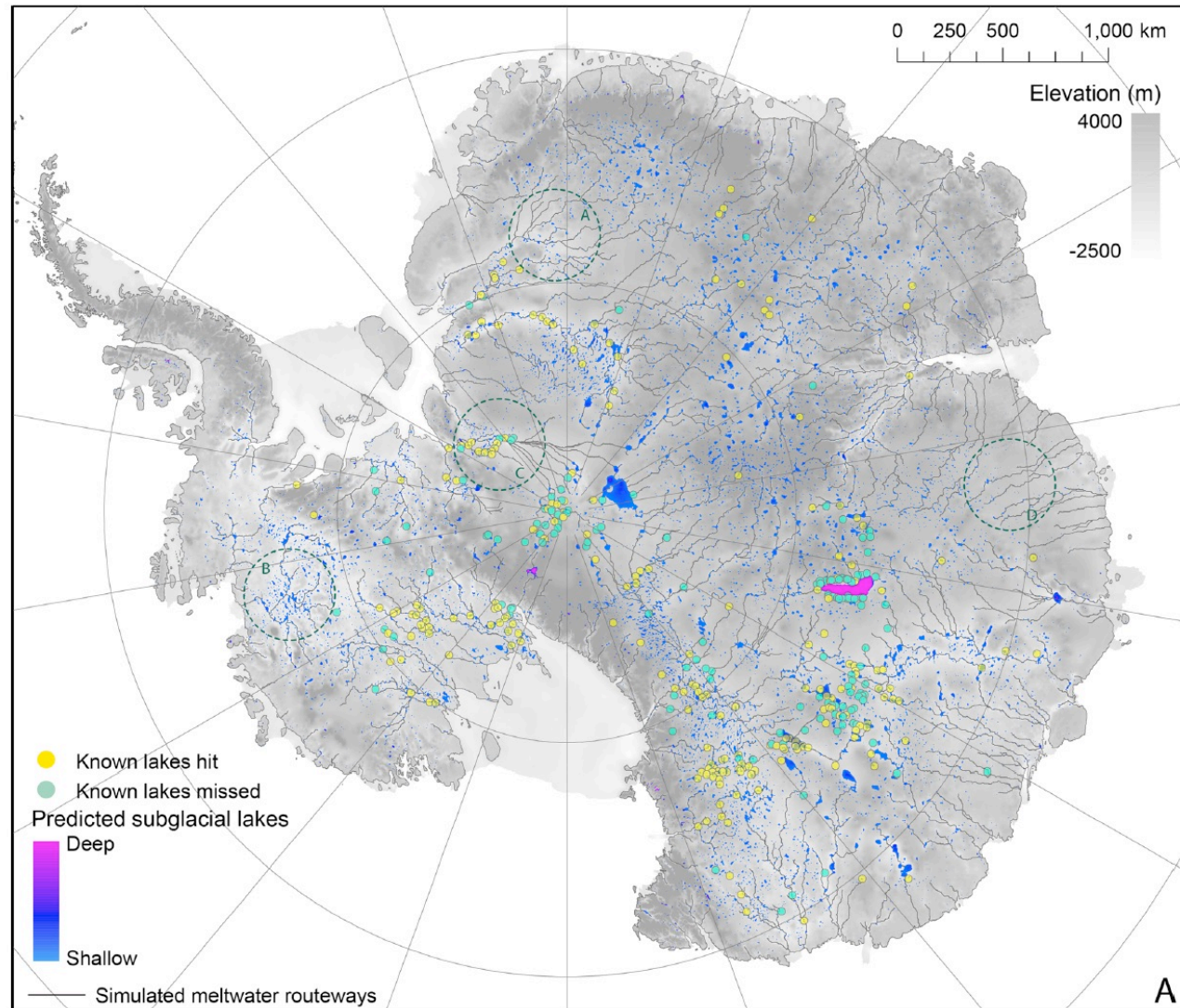
Suckale et al. (2014)

Englacial Antarctica



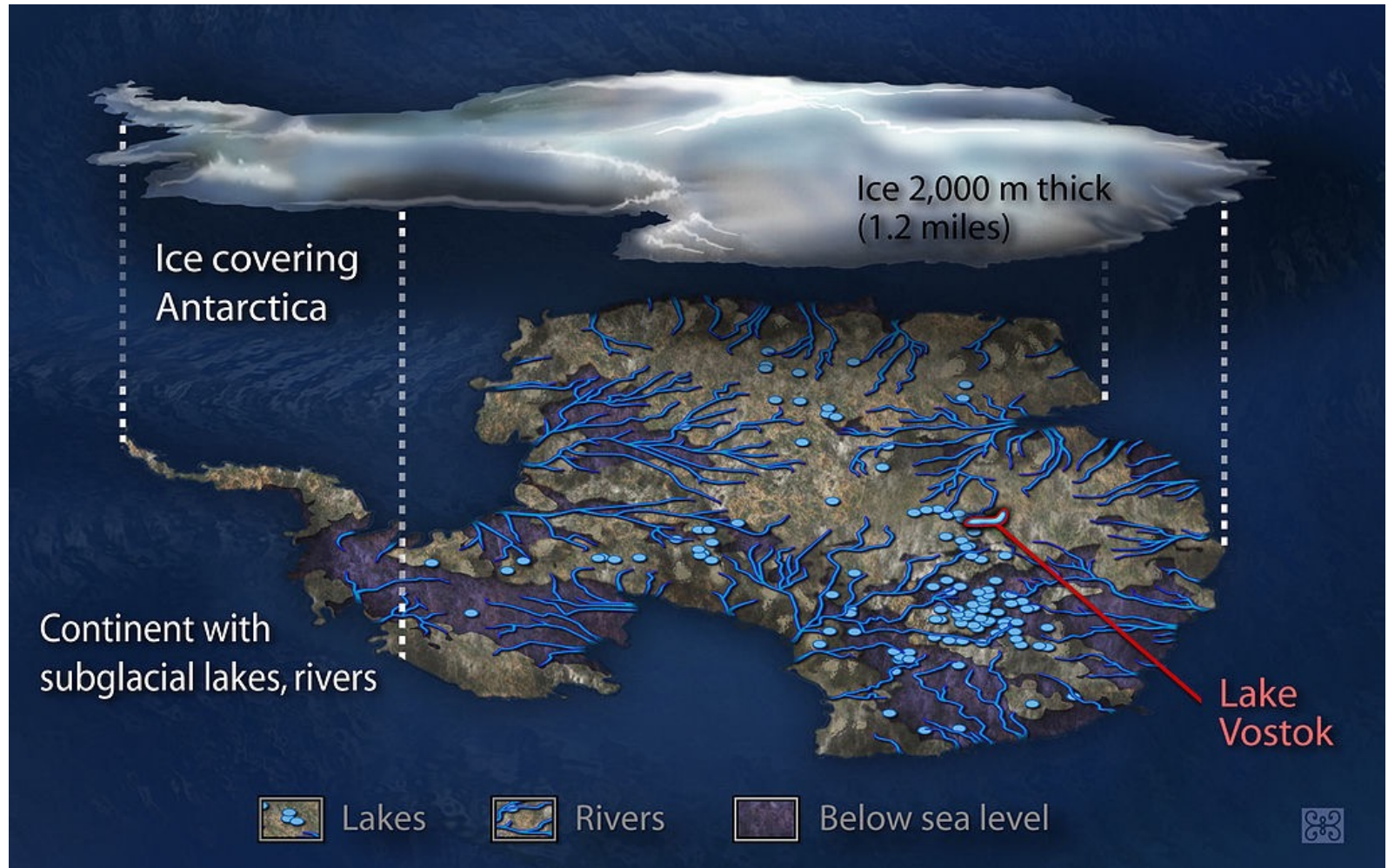
Kuipers Munneke et al. (2014)

Subglacial Antarctica



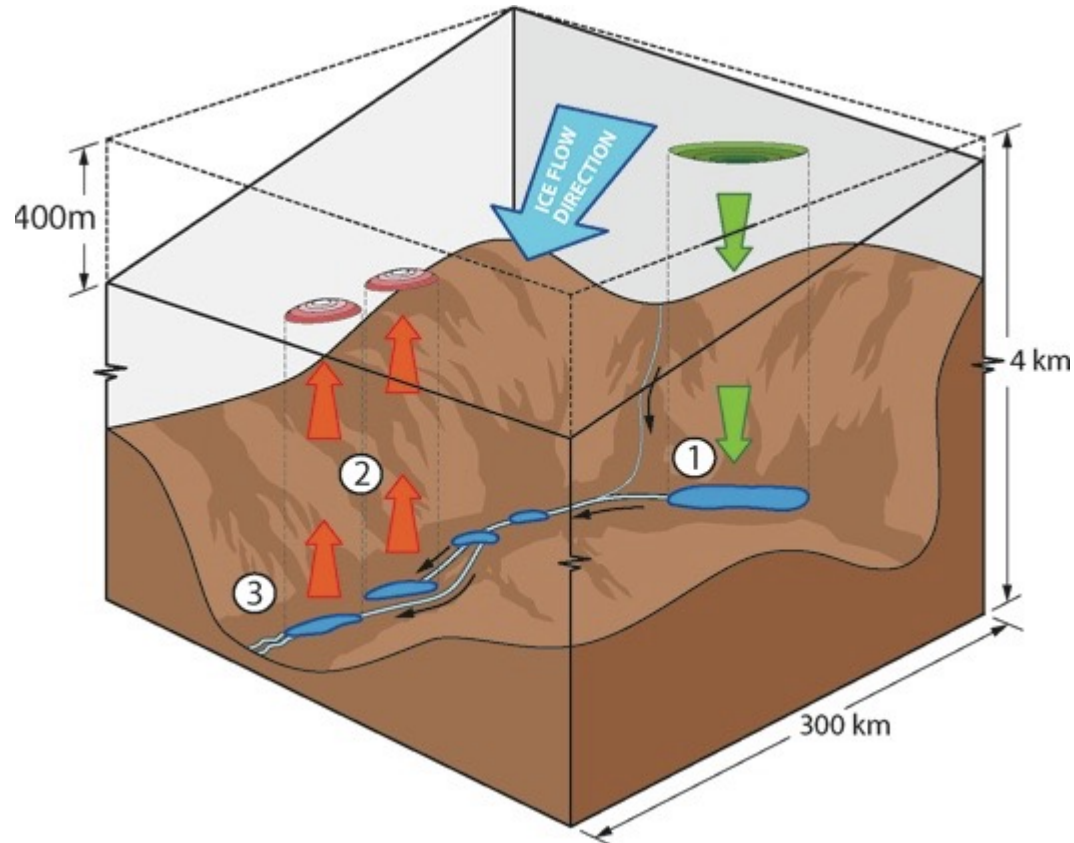
Livingstone et al. (2013)

Subglacial Antarctica



Zina Deretsky / NSF (US National Science Foundation), via Wikimedia Commons

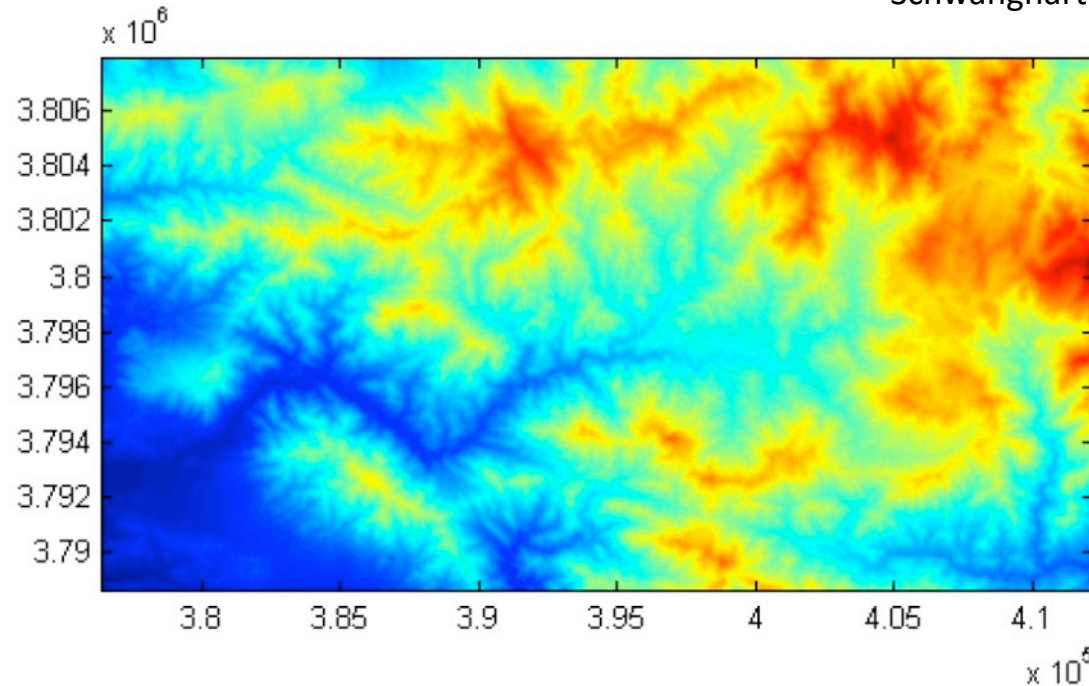
Subglacial Antarctica



Wingham et al. (2006)

How do we determine where water flows subglacially?

Subareally:



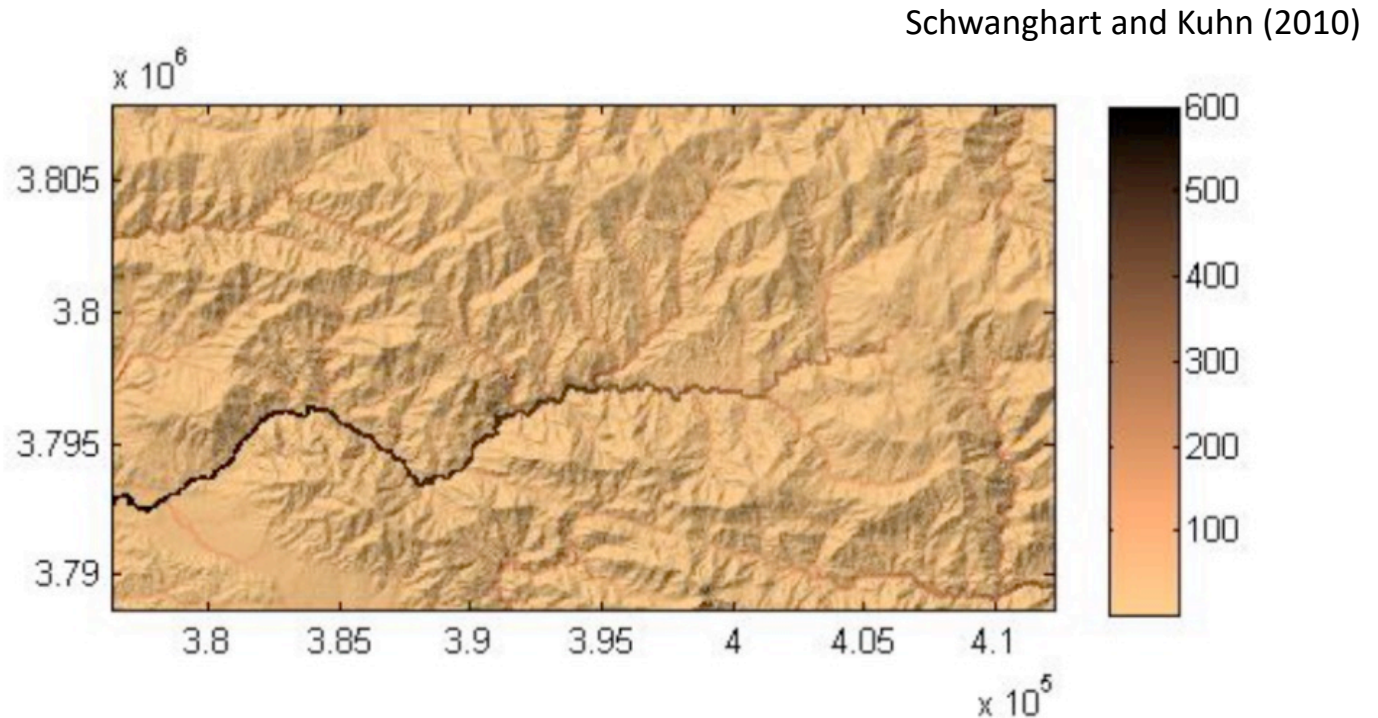
Water flows downhill!

More precisely, down
gravitational potential

$$\Phi = \rho_w g z$$

How do we determine where water flows subglacially?

Subareally:



Water flows downhill!

More precisely, down
gravitational potential

$$\Phi = \rho_w g z$$

How do we determine where water flows subglacially?

Subglacially water can be pressurized, so water pressure p_w is also important

$$\Phi = p_w + \rho_w g z_b$$

Journal of Glaciology, Vol. 11, No. 62, 1972

MOVEMENT OF WATER IN GLACIERS*

By R. L. SHREVE

How do we determine where water flows subglacially?

Subglacially water can be pressurized, so water pressure p_w is also important

$$\Phi = p_w + \rho_w g z_b$$

Assume water pressure equal to overburden pressure

$$\Phi = p_i + \rho_w g z_b$$

Journal of Glaciology, Vol. 11, No. 62, 1972

MOVEMENT OF WATER IN GLACIERS*

By R. L. SHREVE

How do we determine where water flows subglacially?

Subglacially water can be pressurized, so water pressure p_w is also important

$$\Phi = p_w + \rho_w g z_b$$

Assume water pressure equal to overburden pressure

$$\Phi = \rho_i g H + \rho_w g z_b$$

H is ice thickness = $H = z_s - z_b$

Journal of Glaciology, Vol. 11, No. 62, 1972

MOVEMENT OF WATER IN GLACIERS*

By R. L. SHREVE

How do we determine where water flows subglacially?

Subglacially water can be pressurized, so water pressure p_w is also important

$$\Phi = p_w + \rho_w g z_b$$

$$\Phi = \rho_i g (z_s - z_b) + \rho_w g z_b$$

Journal of Glaciology, Vol. 11, No. 62, 1972

MOVEMENT OF WATER IN GLACIERS*

By R. L. SHREVE

How do we determine where water flows subglacially?

Subglacially water can be pressurized, so water pressure p_w is also important

$$\Phi = p_w + \rho_w g z_b$$

Rearrange:

$$\Phi = \rho_i g z_s + (\rho_w - \rho_i) g z_b$$

The ice surface is much more important than the bed for subglacial hydrology!

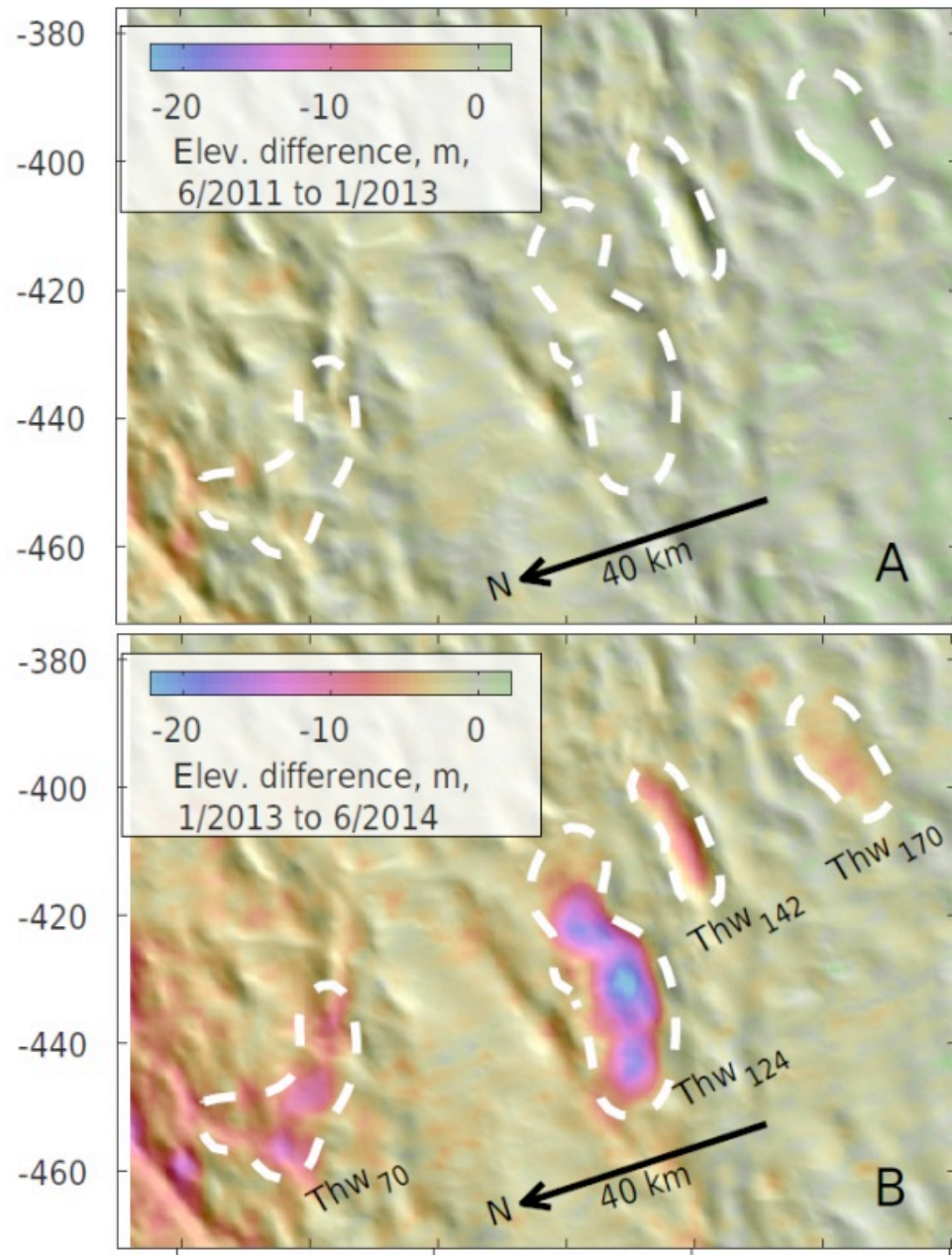
Journal of Glaciology, Vol. 11, No. 62, 1972

MOVEMENT OF WATER IN GLACIERS*

By R. L. SHREVE

Subglacial Antarctica

Smith et al. (2017)



Impacts (Greenland, Antarctica and glaciers):

Impacts (Greenland, Antarctica and glaciers):

- Mass movement
- Albedo
- Ice-shelf calving
- Control input of water into fjords, proglacial environment, sub-shelf cavity.
- Basal sliding
- Downstream water resources
- Erosion
- Proglacial landscaping
- Moves heat around – rheology/viscosity
- Changes humidity over the surface

Summary

Supraglacial: lakes, firn aquifers, streams, buried lakes.

Englacial: moulins, crevasses, shear margins.

Subglacial: channels, water sheets, linked cavities, saturated till, lakes.

These are complex, interconnected hydrological systems!

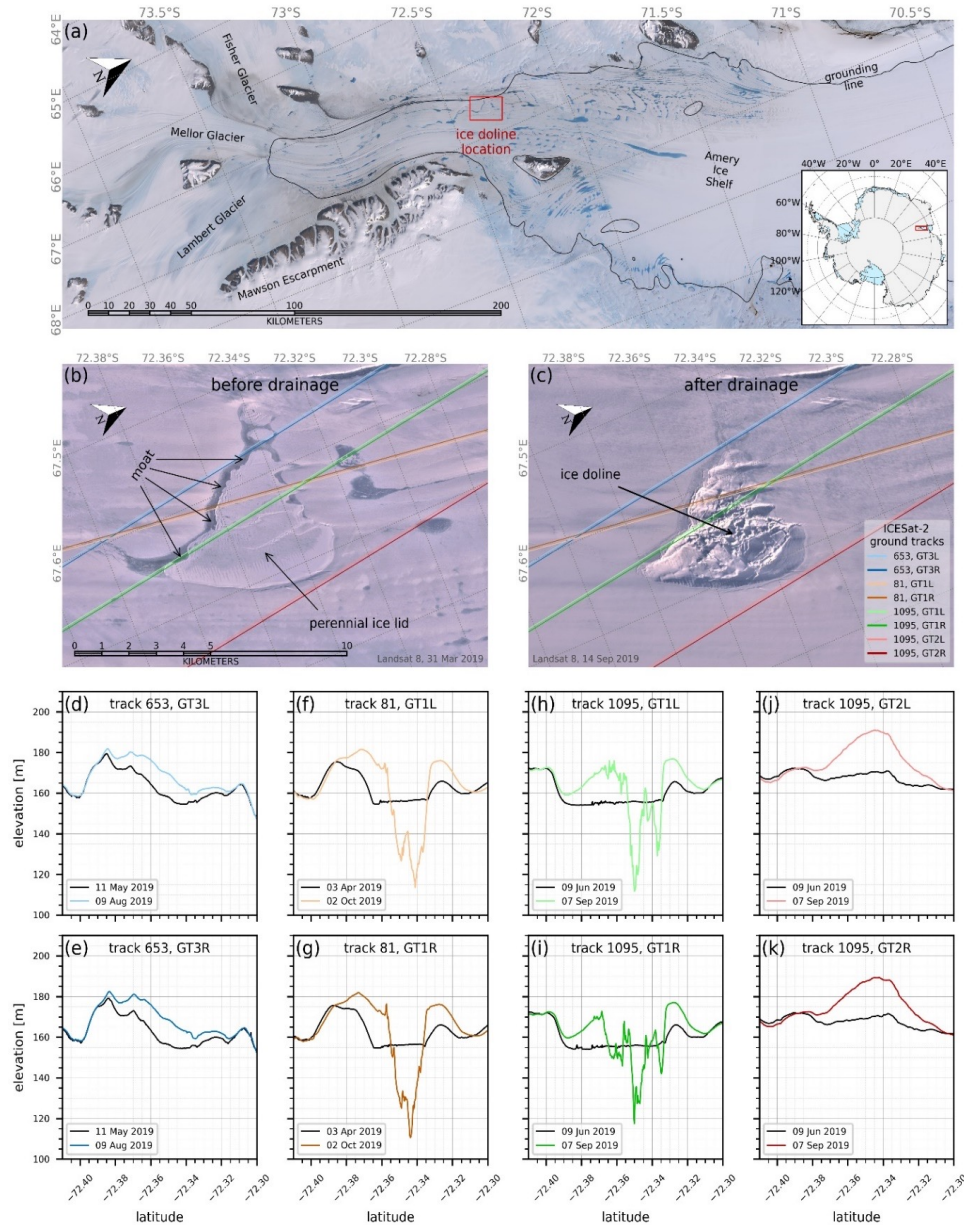
Wide-ranging impacts, from ice dynamics to SMB.

Theory is not well developed and observations of some aspects (englacial, subglacial) is very difficult

References

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